X-SAM: Enhancing Sharpness-Aware Minimization via Principal-Eigenvector-Aligned Gradient Correction

Anonymous Authors[1]

Abstract

Sharpness-Aware Minimization (SAM) aims to improve generalization by minimizing the worst-case loss under small parameter perturbations. However, its optimization behavior during training does not always align with this theoretical expectations, as both sharp and flat regions can exhibit similarly small perturbed losses. Consequently, the optimization trajectory may still drift toward sharp regions, undermining the intended effect of SAM method. To address this issue, we study SAM from a geometric perspective centered on the top eigenvalue of the Hessian. Specifically, we characterize sharpness using the angle between the gradient and the principal eigenvector of the Hessian. Our analysis reveals that when this angle is less than or equal to 90°, the effect of SAM's sharpness regularization can be significantly weakened. Motivated by this insight, we propose principal eigenvector-aligned SAM (X-SAM), which explicitly corrects the gradient via an orthogonal decomposition along the Hessian's principal eigenvector, enabling more direct and efficient regularization of the top Hessian eigenvalue. We provide theoretical guarantees on the convergence of X-SAM, and extensive experimental evaluations demonstrate its practical effectiveness and superior generalization performance.

1. Introduction

Modern deep neural networks are typically heavily overparameterized, resulting in highly non-convex loss landscapes with numerous global minima. Consequently, identifying solutions that generalize well among these minima has therefore become a central challenge in modern machine learning. A substantial body of work has investigated generalization from multiple perspectives (Wilson et al., 2017; Foret et al., 2021; Jastrzębski et al., 2018). From the viewpoint of optimization dynamics, prior studies show that implicit biases of stochastic gradient descent (SGD) and adaptive methods such as Adam (Kingma & Ba, 2015) play a key role in their differing generalization behaviors (Wilson et al., 2017; Soudry et al., 2018). Other work examines the impact of batch size, demonstrating that large-batch training tends to converge to a sharp minimal with inferior generalization (Keskar et al., 2017). In addition, explicit regularization methods that incorporate second-order information have been proposed to improve robustness in parameter space (Liu et al., 2022; Liang et al., 2019; Yao et al., 2020). Despite these advances, existing approaches remain limited in their ability to directly characterize and explain generalization behavior in highly complex loss landscapes, motivating renewed exploration from the perspective of loss geometry.

Against this backdrop, recent studies have shown that the geometric properties of the loss landscape in neural network parameter space—particularly the sharpness of local minima—are closely linked to generalization performance (Keskar et al., 2017; Jiang et al., 2019; Xie et al., 2020). In general, models converging to flatter minima tend to exhibit better generalization, whereas those reaching sharper minima are more susceptible to overfitting and reduced robustness (Hochreiter & Schmidhuber, 1997). This observation has motivated increasing interest in understanding how optimization algorithms navigate the loss landscape and seek flat regions during training. A representative example is Sharpness-Aware Minimization (SAM) (Foret et al., 2021; Kaddour et al., 2022; Yue et al., 2023), which explicitly controls solution sharpness by minimizing the worst-case loss within a local neighborhood of the current parameters.

Intuitively, SAM aims not only to minimize the loss at the current parameter point but also to favor solutions that remain flat regions under small perturbations, thereby encouraging convergence to flatter and more robust minima with better generalization performance. Guided by this principle, SAM has been shown to improve the generalization performance of gradient-based optimizers such as SGD and Adam, achieving notable performance gains across domains, includ-

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

ing computer vision (Foret et al., 2021; Li et al., 2024) and natural language processing (Bahri et al., 2021). However, recent studies indicate that the perturbed loss minimized by SAM can be small at both flat and sharp minima (Zhuang et al., 2022; Luo et al., 2024), suggesting that minimizing the worst-case perturbed loss does not necessarily guarantee truly sharpness-aware solutions. Consequently, SAM may still converge to sharp regions of the loss landscape in practice.

To better understand this phenomenon, we conduct an empirical analysis by training a SimpleCNN model (Jastrzebski et al., 2021) on CIFAR-10 using SAM with SGD, and analyze the angle between the gradient direction and the principal eigenvector of the Hessian throughout training. The top Hessian eigenvalue serves as a measure of sharpness: larger values correspond to sharper regions, while smaller values indicate flatter ones. Our observations reveal a clear geometric pattern: when the angle between the gradient and the principal eigenvector is smaller than 90°, the optimization trajectory tends to move toward sharper regions, whereas angles larger than 90° encourage updates towards flatter regions.

Motivated by this insight, and supported by recent PAC-Bayesian results (Luo et al., 2024) that establish a direct connection between the top Hessian eigenvalue and generalization error, we propose X-SAM, a novel and efficient variant of SAM. X-SAM (i) leverages the principal eigenvector of the Hessian as a reliable indicator of the direction with the largest curvature, and (ii) refines the parameter update by attenuating the gradient component projected onto this eigenvector, thereby explicitly steering optimization toward flatter regions of the loss landscape. Extensive experimental results demonstrate that X-SAM consistently improves generalization performance compared to four state-of-the-art SAM-based methods. The main contributions of this paper are summarized as follows:

- First, we provide a geometric analysis of SAM by examining the angle between the gradient and the Hessian's principal eigenvector, and show that enlarging the angle effectively reduces the top Hessian eigenvalue, leading to improved generalization. Motivated by this observation, we propose a novel variant of SAM, termed X-SAM.
- Second, we theoretically demonstrate that X-SAM can converge to flat regions with low curvature by removing the gradient component aligned with the principal eigenvector, and further prove the convergence of X-SAM under non-convex stochastic optimization.
- Third, we conduct extensive experiments comparing X-SAM with four state-of-the-art SAM-based methods, including SAM (Foret et al., 2021), GSAM (Zhuang et al., 2022), WSAM (Yue et al., 2023), and Eigen-SAM (Luo et al., 2024). The results indicate that X-SAM consistently achieves superior or comparable generalization performance. For example, when training on CIFAR-10, X-SAM improves test accuracy by an average of 1.30% (up to 1.43%), and on CIFAR-100, it achieves an average gain of 2.57% with a maximum improvement of 2.95%.

2. Preliminaries

When optimizing neural networks with gradient-based methods, traditional approaches aim to minimize the empirical training loss $f(w)$ to obtain model parameters w (Drucker & Cun, 1991; Song et al., 2024; Deng, 2012; Netzer et al., 2011). However, as noted by Chaudhari (Chaudhari et al., 2019), such training procedures often converge to sharp regions of the loss landscape, which can severely degrade generalization performance. In contrast, Sharpness-Aware Minimization (SAM) seeks solutions located in low-loss region that are robust to small parameter perturbations (Ahn et al., 2023; Barrett & Dherin, 2020). Formally, SAM formulates the optimization problem as

$$\min_w f_p(w) \triangleq \min_w \Big(\max_{\|\epsilon\| \le \rho} f(w+\rho\epsilon) \Big), \tag{1}$$

where ρ is a predefined hyperparameter controlling the perturbation radius, and $f_p(w)$ denotes the maximum value of $f(w)$ within the corresponding neighborhood. By parameterizing the perturbation as $\delta = \rho\epsilon$ (with $\|\epsilon\| \le 1$), and assuming ρ is sufficiently small, we apply a first-order Taylor expansion of $f(w+\delta)$ around w. Under this approximation, the inner maximization problem in Eq. (1) admits the following solution:

$$\begin{aligned} \arg\max_{\|\delta\|\le\rho} f(w+\delta) &\approx \arg\max_{\|\delta\|\le\rho} \big(f(w) + \delta^\top \nabla f(w) + O(\rho^2) \big) \\ &= \arg\max_{\|\delta\|\le\rho} \delta^\top \nabla f(w) \\ &= \rho \frac{\nabla f(w)}{\|\nabla f(w)\|}. \end{aligned} \tag{2}$$

As a result, the optimization problem of SAM reduces to

$$\min_w f\left(w + \rho \frac{\nabla f(w)}{\|\nabla f(w)\|} \right). \tag{3}$$

In the stochastic setting, given a mini-batch of size β_k and the learning rate η, the parameter update for SAM is formulated as

$$w_{k+1} = w_k - \eta \nabla f_{\beta_k} \left(w_k + \rho \frac{\nabla f_{\beta_k}(w_k)}{\|\nabla f_{\beta_k}(w_k)\|_2} \right). \tag{4}$$

The surrogate minimization problem shown in Eq. (3) is solved by performing stochastic gradient descent along the

corresponding negative gradient direction. Overall, SAM optimizes the parameters with respect to the perturbed loss surface $f_p(w)$, rather than directly minimizing the original loss $f(w)$, thereby biasing the optimization toward flatter regions of the loss landscape.

3. Methodology

3.1. Empirical Analysis

In this subsection, we empirically analyze parameter update directions during training by examining the angle between the gradient and the principal eigenvector of the Hessian. We denote this angle as $\theta_{\langle g_t, v_1\rangle}$, which reflects whether the updates are aligned with directions of maximum curvature in the loss landscape.

Specifically, we train a six-layer SimpleCNN model (Jastrzebski et al., 2021) on CIFAR-10 using SAM with SGD, and monitor $\theta_{\langle g_t, v_1\rangle}$ throughout training. After obtaining the gradient from the second backward pass and estimating the Hessian's principal eigenvector, we flatten and ℓ_2-normalize the gradient tensor of each layer and concatenate them to form the full-model gradient vector. The principal eigenvector of each layer is processed similarly. The angle $\theta_{\langle g_t, v_1\rangle}$ is then computed as the arccosine of the cosine similarity between the two full-model vectors. To capture the optimization dynamics, we record $\theta_{\langle g_t, v_1\rangle}$ every five iterations. The resulting statistics are visualized in Fig. 1 (blue bars), where the x-axis denotes the angle (in degrees) and the y-axis represents the frequency of occurrence.

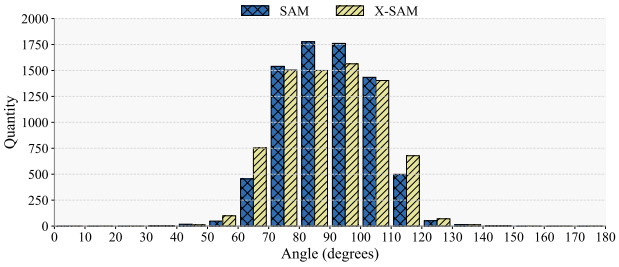


Figure 1. Angle statistics of a 6-layer SimpleCNN trained on CIFAR-10. The figure illustrates the distribution of the angles between gradients (calculated from the second forward-backward pass of SAM and X-SAM) and the principal eigenvectors.

The distribution patterns shown by the blue bars indicate that for SAM, approximately 87% of the angles lie between 70° and 110°. We now analyze the impact of the angle $\theta_{\langle g_t, v_1\rangle}$ on the loss landscape, considering three distinct cases under three idealized assumptions.

First, we consider the case when the angle is less than 70°, simplifying it to the scenario where the gradient is perfectly aligned with the principal eigenvector, i.e., $g_t = \alpha v_1$ with $\alpha > 0$. In this case, the standard gradient descent update is given by $\Delta w = -\eta g_t = -\eta\alpha v_1$. Based on the conclusion drawn in (Damian et al., 2022), the variation of the top eigenvalue along the direction of the principal eigenvector can be calculated approximately as

$$\Delta\lambda_{max} \approx -\eta\alpha\langle\nabla\lambda_{max}, v_1\rangle = -\eta\alpha\sum_{k=1}^{d}(v_1)_k\left(v_1^\top\frac{\partial H}{\partial w_k}v_1\right). \tag{5}$$

Considering that $\eta > 0$, $\alpha > 0$, and $\langle\nabla\lambda_{max}, v_1\rangle < 0$, Eq. (5) implies a positive change in the top Hessian eigenvalue, i.e., $\Delta\lambda_{max} > 0$.

Second, when the gradient is approximately orthogonal to the principal eigenvector v_1, i.e., $\langle g_t, v_1\rangle = 0$, The gradient g_t is decomposed as a linear combination of other eigenvectors $\{v_2, \ldots, v_d\}$: $g_t \approx \sum_{i=2}^{d}\beta_i v_i$. The change in the top eigenvalue is:

$$\Delta\lambda_{max} \approx -\eta\sum_{i=2}^{d}\beta_i\langle\nabla\lambda_{max}, v_i\rangle = -\eta\sum_{i=2}^{d}\beta_i\left(v_1^\top[\nabla H\cdot v_i]v_1\right). \tag{6}$$

When the gradient g_t is orthogonal to the principal eigenvector v_1 and the gradient of the eigenvalue $\nabla\lambda_1$ is highly collinear with v_1, the update of the top Hessian eigenvalue is approximately zero.

Third, we consider the case where the gradient g_t forms an obtuse angle with the principal eigenvector v_1 (i.e., $\langle g_t, v_1\rangle < 0$). To analyze the update process of the eigenvalue, we decompose the gradient into a component parallel to v_1 and an orthogonal component $g_\perp$: $g_t = |g_t|\cos\theta_{\langle g_t, v_1\rangle}\cdot v_1 + g_\perp$,where θ denotes the angle between g_t and v_1. Consequently, the first-order approximation of the change in the top eigenvalue λ_1 can be derived as:

$$\begin{aligned}\Delta\lambda_{max} &\approx \langle\nabla\lambda_{max}, -\eta g_t\rangle \\ &\approx -\eta\langle\nabla\lambda_{max}, |g_t|\cos\theta_{\langle g_t, v_1\rangle}\cdot v_1 + g_\perp\rangle.\end{aligned} \tag{7}$$

Focusing on the dominant principal eigenvalue and ignoring orthogonal drift, we assume $\langle\nabla\lambda_{max}, v_1\rangle < 0$. Consequently, an obtuse update angle $\theta_{\langle g_t, v_1\rangle}$ ($\cos\theta < 0$) results in a reduction of λ_1.

Based on these mathematical derivations and empirical observations, we summarize the evolutionary trends of the eigenvalue into three distinct dynamic regimes:

Acute Regime (Curvature Accumulation). For $\theta_{\langle g_t, v_1\rangle} < 70°$, the gradient aligns with the principal curvature direction. Consistent with Eq. (5), this alignment induces a positive drift in the maximum eigenvalue ($\Delta\lambda_{\max} > 0$), creating a "curvature accumulation zone" where the landscape steepens significantly.

Orthogonal Regime (Optimization Blind Spot). When the gradient is nearly orthogonal to the principal eigenvector ($\theta_{\langle g_t, v_1\rangle} \approx 90°$), the projection term dominates.

As indicated by Eq. (6), this orthogonality ($\langle g_t, v_1 \rangle \approx 0$) leads to negligible first-order changes in the eigenvalue ($\Delta\lambda_{\max} \approx 0$), regardless of the gradient magnitude. This theoretically characterizes an "optimization blind spot," where the optimizer fails to suppress local sharpness.

Obtuse Regime (Sharpness Reduction). For angles $\theta_{\langle g_t, v_1 \rangle} > 110°$, the dynamics are governed by the negative cosine term in Eq. (7). This generates a negative increment ($\Delta\lambda_{\max} < 0$), effectively penalizing the top eigenvalue. Consequently, updates in this regime serve as the primary mechanism for steering the trajectory toward flatter and more robust minima.

3.2. The X-SAM Algorithm

Building on the preceding analysis of how the gradient–principal eigenvector angle influences the top Hessian eigenvalues, the main idea of X-SAM is to actively enlarge the angle $\theta_{\langle g_t, v_1 \rangle}$. By doing so, X-SAM suppresses updates along high-curvature directions (i.e., the principal eigenvector of Hessian), steering the optimization trajectory toward flatter and more robust minima. To achieve this objective, we propose an enhancement to the standard SAM method via the use of an explicit gradient correction. This new variant, named X-SAM, is outlined in Algorithm 1 when using SGD as the base optimizer. At the beginning of each iteration, X-SAM first estimates the principal eigenvector with the Power Iteration method (Luo et al., 2024) (as shown in Lines 3-10). Then, the gradient-based perturbation is computed and added to the current model weights to generate the perturbed parameters (as shown in Lines 11-12).

Subsequently, in Line 13, X-SAM normalizes the gradient at the perturbed position as follows:

$$\hat{\nabla} f(w^{adv}) = \frac{\nabla f(w^{adv})}{\|\nabla f(w^{adv})\|}. \tag{8}$$

This operation removes the influence of gradient magnitude on directional judgment, allowing the method to focus primarily on directional information.

Next, as shown in Line 14 of Algorithm 1, X-SAM decomposes the normalized gradient into two orthogonal components: one parallel to the principal eigenvector (representing the sharpest direction) and the other orthogonal to it.

$$\hat{\nabla} f(w^{adv}) = \nabla f(w^{adv})\| + \nabla f(w^{adv})\perp. \tag{9}$$

Finally, as shown in Line 15 of Algorithm 1, X-SAM further corrects the gradient direction based on the decomposition by Eq. (9). By introducing a hyperparameter α to attenuate the component along the principal eigenvector, X-SAM artificially increases the angle between the final update gradient and the principal eigenvector.

$$\begin{aligned}\nabla f(w)_{\text{X-SAM}} &= \hat{\nabla} f(w^{adv}) \\ &\quad - \alpha \cdot \operatorname{sign}(\langle \nabla f(w^{adv}), v \rangle) \nabla f(w^{adv})_{\|}.\end{aligned} \tag{10}$$

where $\operatorname{sign}(\langle \nabla f(w^{adv}), v \rangle)$ is incorporated to resolve the directional ambiguity of the eigenvector. Crucially, this term ensures that the correction always acts to enlarge the angle $\theta_{\langle g_t, v_1 \rangle}$, regardless of whether the initial gradient and the eigenvector form an acute or obtuse angle. This operation suppresses updates along the sharpest direction while preserving the descent component along flatter directions.

By introducing fine-grained control over the gradient direction, X-SAM significantly alters the angle distribution between the gradient and the principal eigenvector, as illustrated by the yellow boxes in Figure 1. Specifically, it breaks the high concentration of gradients around the orthogonal state (90°) and increases the proportion of both acute and obtuse angles. This distributional shift indicates that X-SAM actively reshapes the update trajectory, thereby achieving a direct constraint on the top eigenvalue of the Hessian and addressing the inherent limitations of standard SAM. Consequently, this mechanism leads to a marked enhancement in the model's generalization performance.

This realignment of the update trajectory is directly mapped to the final convergence regions within the loss landscape (see Figure 2). Experimental comparisons reveal that baseline optimizers, such as SGD and Adam, are highly susceptible to becoming trapped in sharp minima characterized by high local curvature (Keskar et al., 2017).

While standard SAM attempts to seek flatter regions, it exhibits limitations, often stalling in suboptimal saddle point areas due to a lack of directional precision. In contrast, X-SAM overcomes this by leveraging its core principal eigenvector-corrected gradient mechanism. This approach actively identifies and suppresses update weights along sharp directions, enabling the model to successfully bypass high-curvature traps and precisely anchor within the flatter and broader minima of the loss landscape. Consequently, by effectively suppressing the top Hessian eigenvalue and optimizing the convergence path, X-SAM ensures superior generalization performance and enhanced model robustness.

4. Theoretical Analysis

4.1. Reducing Top Eigenvalue Enhances Generalization

Assumption 1. *The objective function $f : \mathbb{R}^d \to \mathbb{R}$ is four times continuously differentiable (C^4) in the region of interest. Furthermore, its third and fourth derivatives are uniformly bounded by constants $M_3, M_4 \geq 0$: $\|\nabla^3 f(x)\|_{\text{op}} \leq M_3, \|\nabla^4 f(x)\|_{\text{op}} \leq M_4$.*

Algorithm 1 X-SAM Algorithm

Require: Network f_w with parameters w; Learning rate $\eta > 0$; Neighborhood size $\rho > 0$; Total iterations T; Hessian update interval p; Power iteration steps q.

for $t = 1, 2, \ldots, T$ **do**
  Compute mini-batch loss $f(w)$
  if $t \bmod p = 1$ **then**
    Initialize $\hat{v}$ as a random unit vector
    for $s = 1, 2, \ldots, q$ **do**
      Compute Hessian-vector product: $\hat{v} = \nabla^2 f(w) \cdot \hat{v}$
      Normalize $\hat{v}$: $\hat{v} \leftarrow \hat{v}/\|\hat{v}\|$
    end for
    let $v_1 = \hat{v}$
  end if
  Compute the perturbation: $\epsilon_t^{\text{SAM}} = \rho \frac{\nabla f(w_t)}{\|\nabla f(w_t)\|}$
  Perturb the model parameters: $w_t^{\text{adv}} = w_t + \epsilon_t^{\text{SAM}}$
  Normalize the gradient:
$$\hat{\nabla} f(w_t^{\text{adv}}) = \frac{\nabla f(w_t^{\text{adv}})}{\|\nabla f(w_t^{\text{adv}})\|}$$
  Decompose the gradient along the principal eigenvector:
$$\hat{\nabla} f(w_t^{\text{adv}}) = \nabla f(w_t^{\text{adv}})_{\parallel} + \nabla f(w_t^{\text{adv}})_{\perp}$$
  Perform gradient correction:
$$\nabla f(w_t)_{\text{X-SAM}} = \hat{\nabla} f(w_t^{\text{adv}}) - \alpha \cdot \text{sign}(\langle \nabla f(w_t^{\text{adv}}), v_1 \rangle) \nabla f(w_t^{\text{adv}})_{\parallel}$$
  Update parameters: $w_{t+1} = w_t - \eta \nabla f(w_t)_{\text{X-SAM}}$
end for

Assumption 2. *For the Hessian matrix $H(x) := \nabla^2 f(x)$, the largest eigenvalue $\lambda_1(x)$ is simple. Specifically, there exists a strict eigengap $\gamma > 0$ separating the largest eigenvalue from the rest of the spectrum: $\lambda_1(x) - \lambda_2(x) \geq \gamma > 0$.*

Assumption 3. *The step size η is chosen sufficiently small such that the geometry of the Hessian remains stable. Specifically, we assume the perturbation magnitude satisfies: $\|\nabla^2 f(x + \Delta x) - \nabla^2 f(x)\|_{\text{op}} < \frac{\gamma}{2}$.*

Theorem 4.1 (Decrease of the leading eigenvalue and Orthogonality and Sharpness Reduction)**.** *Suppose Assumptions 1-3 hold. Define $s := \|\Delta H\|_{\text{op}}$, and assume $s < \gamma$. Assume the uniform third-derivative bound $\left|\nabla^3 f(z)[a, b, c]\right| \leq M_3 \|a\| \, \|b\| \, \|c\|$, and $\Delta x = \eta \, \tilde{h}$. Moreover, the top eigenvalue satisfies*

$$\lambda_{max}(H + \Delta H) \leq \lambda_{max}(H) + v_1^{\top} \Delta H v_1 + \frac{4(\|H\|_{\text{op}} + s)}{\gamma - s} s \tag{11}$$

This result establishes a sufficient condition that ensures that the corrected update direction induces a local flattening effect on the loss surface, reducing the spectral sharpness. We provide detailed proof in Appendix A.

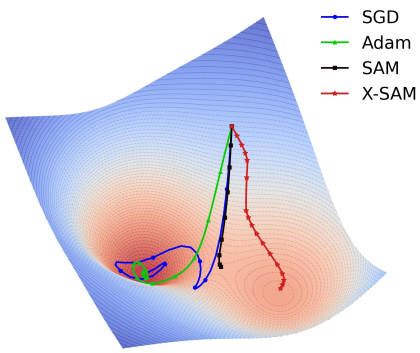


Figure 2. Comparison of X-SAM with SGD, Adam, and SAM. The visualization illustrates that X-SAM converges to flatter minima than the baseline optimizers.

4.2. Convergence Analysis

Assumption 4. *The objective function $f : \mathbb{R}^d \to \mathbb{R}$ is β-smooth; that is, its gradient ∇f satisfies the β-Lipschitz continuity condition.*

Assumption 5. *The stochastic gradient is conditionally unbiased with bounded conditional variance, i.e., $\mathbb{E}[\hat{h}_t \mid x_t] = h_t$, and $\mathbb{E}[\|\hat{h}_t - h_t\|^2 \mid x_t] \leq \sigma^2$.*

Assumption 6. *There exists a constant $C_1 > 0$ such that the perturbation term ϵ_t satisfies $\|\epsilon_t\|^2 \leq 1 + C_1 \alpha^2$ almost surely.*

Theorem 4.2 (Convergence of X-SAM with eigen-projection (nonconvex))**.** *Suppose Assumptions 4-6 hold. Consider a non-convex function f that is β-smooth (i.e., ∇f is β-Lipschitz) and bounded from below by $f^\star$. Suppose we access a stochastic estimator $\hat{h}_t$ of the perturbed gradient $h_t = \nabla f(x_t + \rho \epsilon_t)$ satisfying unbiasedness and bounded variance σ^2. With learning rate $\eta = \min\left\{\frac{1}{2\beta}, \sqrt{\frac{\Delta}{\beta \sigma^2 T}}\right\}$ and $\Delta := f(w_0) - f_*$, we have*

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\|\nabla f(x_t)\|^2 \leq c_1 \left(\frac{\beta \Delta}{T} + \sqrt{\frac{\beta \Delta \sigma^2}{T}} \right) + c_2 (1 + 2\alpha^2) D_\rho \tag{12}$$

where $D_\rho = \beta^2 \rho^2 (1 + C_1 \alpha^2)$ represents the bias induced by the perturbation radius ρ, and c_1, c_2 are some constants.

Theorem 4.2 implies that the X-SAM converges at a rate of $O(1/\sqrt{T})$ for non-convex stochastic optimization, matching the standard convergence rate of first-order stochastic gradient optimizers like SGD, up to a perturbation bias term $O(\rho^2)$. The detailed proof can be found in Appendix B.

5. Experiments

5.1. Experimental Settings

Experiments were conducted on a Linux-based server equipped with NVIDIA RTX 4090 GPUs. The software

Table 1. Summarization of Test Accuracy (%) results. The best accuracy results are highlighted in bold.

Datasets	Methods	Test Accuracy (%)				
		ResNet-18	ResNet-50	WideResNet	AlexNet	Avg.
CIFAR-10	SAM	93.64 ± 0.30	92.33 ± 0.54	95.01 ± 0.18	88.66 ± 0.56	92.41
	WSAM	93.01 ± 0.42	91.68 ± 0.25	94.23 ± 0.08	90.08 ± 0.35	92.25
	GSAM	93.80 ± 0.24	92.16 ± 0.39	94.67 ± 0.42	88.59 ± 0.13	92.31
	Eigen-SAM	93.82 ± 0.17	92.57 ± 0.29	95.11 ± 0.10	88.79 ± 0.19	92.57
	X-SAM	$\mathbf{94.71 \pm 0.20}$	$\mathbf{93.76 \pm 0.07}$	$\mathbf{95.53 \pm 0.26}$	$\mathbf{90.73 \pm 0.09}$	**93.68**
CIFAR-100	SAM	74.00 ± 0.24	72.01 ± 1.67	77.96 ± 0.15	60.26 ± 0.35	71.06
	WSAM	73.65 ± 0.01	71.04 ± 0.60	77.19 ± 0.53	65.90 ± 0.36	71.95
	GSAM	73.55 ± 0.09	71.61 ± 0.54	78.15 ± 0.14	60.14 ± 0.21	70.86
	Eigen-SAM	73.74 ± 0.28	71.90 ± 1.07	78.20 ± 0.02	60.49 ± 0.38	71.08
	X-SAM	$\mathbf{74.10 \pm 0.37}$	$\mathbf{74.08 \pm 0.23}$	$\mathbf{78.23 \pm 0.36}$	$\mathbf{68.82 \pm 0.01}$	**73.81**
Fashion-MNIST	SAM	95.22 ± 0.14	94.46 ± 0.44	95.56 ± 0.10	94.34 ± 0.03	94.90
	WSAM	95.32 ± 0.35	94.57 ± 0.13	95.40 ± 0.07	—	95.10
	GSAM	95.39 ± 0.03	94.94 ± 0.20	$\mathbf{95.68 \pm 0.15}$	94.61 ± 0.08	95.16
	Eigen-SAM	95.36 ± 0.24	94.76 ± 0.12	95.61 ± 0.05	—	**95.24**
	X-SAM	$\mathbf{95.52 \pm 0.14}$	$\mathbf{95.16 \pm 0.04}$	95.53 ± 0.04	$\mathbf{94.75 \pm 0.72}$	**95.24**

The symbol "–" indicates that the model failed to converge.

environment consisted of Python 3.8 and PyTorch 2.0.0 (CUDA 11.8). We mainly compared X-SAM with four state-of-the-art SAM-based methods, including SAM (Foret et al., 2021), Eigen-SAM (Luo et al., 2024), WSAM (Yue et al., 2023), and GSAM (Zhuang et al., 2022).

We evaluated each method on CIFAR-10, CIFAR-100, and Fashion-MNIST (Xiao et al., 2017) datasets, with ResNet-18, ResNet-50 (He et al., 2016), AlexNet (Krizhevsky et al., 2017), and WideResNet (Zagoruyko & Komodakis, 2016) as backbone architectures. Standard data augmentations—including random horizontal flipping, 4-pixel padding, and random cropping—were applied.

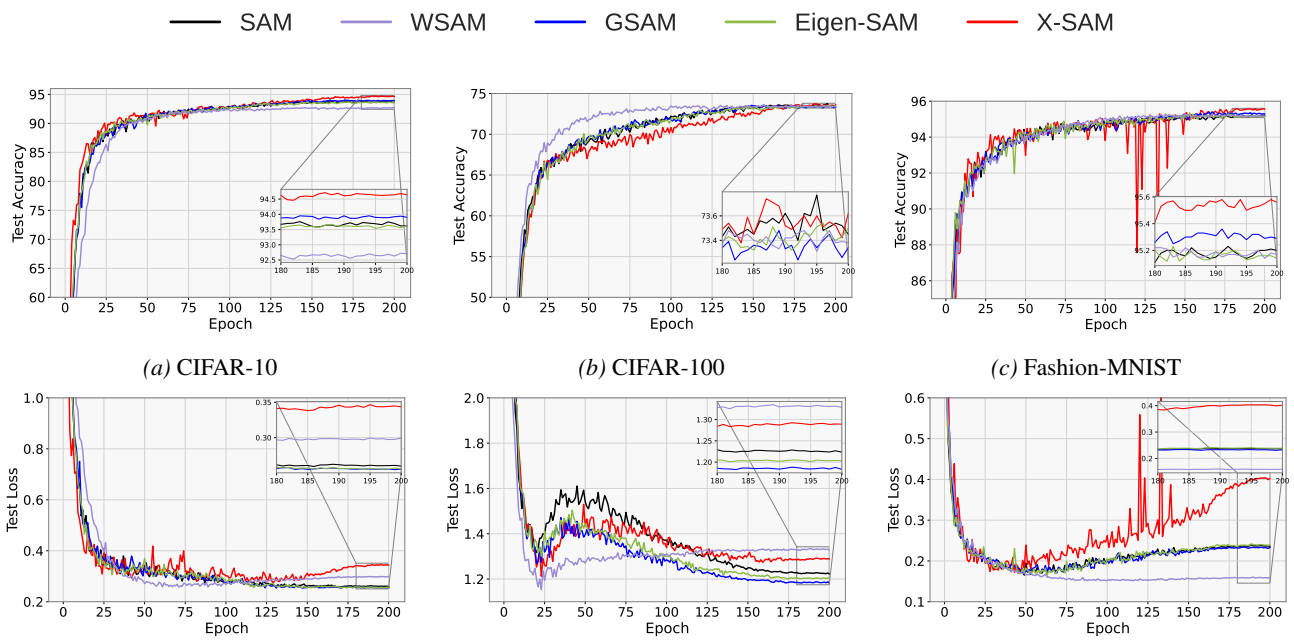
For all methods, we adopted SGD as the base optimizer, with an initial learning rate of 0.1 dynamically adjusted by a cosine schedule, weight decay of 5×10^{-5}, and momentum of 0.9. Models were trained for 200 epochs with a batch size of 256. The perturbation radii were set as follows: SAM and Eigen-SAM used 0.05 for CIFAR-10 and Fashion-MNIST, and 0.1 for CIFAR-100; WSAM used 2 for CIFAR-10 and Fashion-MNIST, and 4 for CIFAR-100; GSAM, originally designed for fine-tuning, followed the same radii as SAM; X-SAM used the same radius as SAM with $\alpha = 0.2$. Each experiment was independently repeated three times to ensure the reliability and stability of the results. We provide detailed hyperparameter settings in Tables C.4 and 2 in Appendix C.

5.2. Performance Comparisons

Table 1 summarizes the test accuracy of X-SAM in comparison with SAM and its variants across different architectures and datasets, while Figure 3 illustrates the learning curves for training ResNet-18 on CIFAR-10, CIFAR-100, and Fashion-MNIST. Based on these experimental results, we draw the following conclusions.

First, X-SAM demonstrates superior generalization performance compared to its counterparts. As shown in Table 1, on CIFAR-10, X-SAM improves test accuracy by 1.27%, 1.43%, 1.37% and 1.11% compared to SAM, WSAM, GSAM, and Eigen-SAM, respectively. On the more challenging CIFAR-100 dataset, this advantage becomes even more pronounced. Excluding the exceptionally strong performance of AlexNet, X-SAM outperforms the four competing methods by 0.81%, 1.51%, 1.03%, and 0.86%, respectively. For the AlexNet architecture alone, the performance gains are substantially larger, reaching 8.56%, 2.92%, 8.68%, and 8.33% over the corresponding baselines. On the relatively simpler Fashion-MNIST dataset, X-SAM continues to exhibit stable superiority, achieving improvements of 0.34%, 0.14%, 0.08%, and 0.00% over SAM, WSAM, GSAM, and Eigen-SAM, respectively.

Second, as illustrated in Figures 3a, 3b, and 3c, X-SAM exhibits superior training efficiency and stability while maintaining a convergence trend comparable to other methods. In particular, the learning curves of X-SAM consistently



(a) CIFAR-10 *(b)* CIFAR-100 *(c)* Fashion-MNIST

(d) CIFAR-10 *(e)* CIFAR-100 *(f)* Fashion-MNIST

Figure 3. Learning curves for training ResNet-18 on CIFAR-10, CIFAR-100, and Fashion-MNIST. Figs 3d, 3e and 3f: Test loss vs. Epochs; Figs 3a, 3b and 3c: Test accuracy vs. Epochs.

remain at higher levels during the mid-to-late stages of training and ultimately achieve the highest test accuracy. This behavior indicates that X-SAM can continuously and reliably guide the optimization trajectory toward parameter regions with improved generalization, rather than getting trapped in local oscillations.

Third, although the test loss of X-SAM (Figures 3d, 3e, and 3f) is not always the lowest, the relationship between its loss and final accuracy highlights its core advantage. While the loss curves of X-SAM often exhibit fluctuations after convergence, the corresponding test accuracy continues to improve and remains consistently high. This behavior indicates that X-SAM's optimization objective is not merely to minimize empirical loss, but to strategically guide gradient updates, steering model parameters toward regions of the loss landscape that are flatter and less sensitive to perturbations, even when the loss value itself is not minimized to the absolute lowest point.

5.3. Evaluations of Top Hessian Eigenvalues

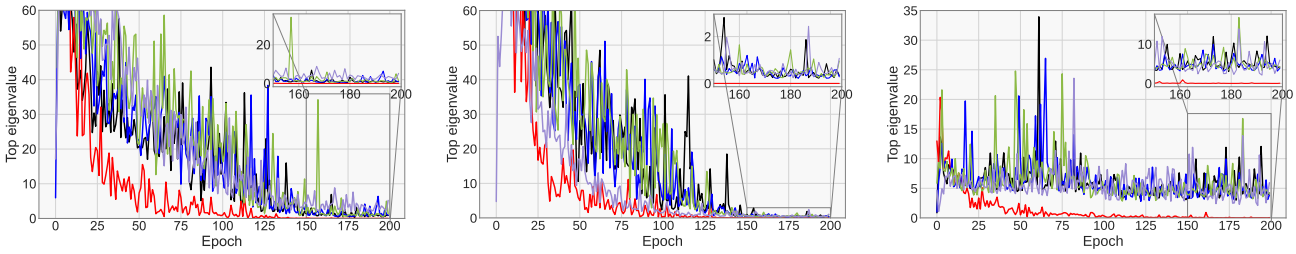
Figure 4 shows the learning curves of the top Hessian eigenvalues over epochs, where the eigenvalues are computed using the Rayleigh quotient method. Specifically, for the Hessian matrix H, the top eigenvalue $\lambda_{\max}$ is obtained from its corresponding principal eigenvector v_1 as $\lambda_{max} = v_1^\top H v_1$.

It can be observed that X-SAM demonstrates clear optimization advantages across the three network architectures: ResNet-18, WideResNet, and AlexNet. Specifically, in the early stages of training, X-SAM reduces the largest Hessian eigenvalue more rapidly and with significantly smaller fluctuations. This behavior suggests that by consistently maintaining lower curvature throughout the optimization process, X-SAM effectively steers the model toward flatter regions of the loss landscape. Consequently, this improves the model's robustness against parameter perturbations and enhances its generalization performance, thereby validating the effectiveness of the proposal.

6. Related Work

Sharpness-Aware Minimization (SAM) is an optimization algorithm designed to improve the generalization of deep neural networks by favoring solutions located in flat regions of the loss landscape (Foret et al., 2021). Since its introduction, SAM has attracted considerable attention, leading to a growing body of work on its theoretical analysis, practical implementations, and extensions (Zhuang et al., 2022; Yue et al., 2023; Li et al., 2024; Luo et al., 2024).

Extensions of SAM. Various extensions of SAM (Foret et al., 2021) have been proposed to improve its optimization behavior and computational efficiency. For example, GSAM (Zhuang et al., 2022) adopts a gradient projection strategy to increase the loss at the original parameter point while keeping the perturbed loss unchanged, thereby implicitly emphasizing the difference between the perturbed



(a) ResNet-18 *(b)* WideResNet *(c)* AlexNet

Figure 4. Learning curves of top Hessian eigenvalues vs. epochs for training ResNet-18, WideResNet, and AlexNet on CIFAR-10.

and unperturbed losses. WSAM (Yue et al., 2023) introduces a weighted combination of the empirical loss and the sharpness measure, effectively integrating sharpness as a explicit regularization term in the optimization objective. Moreover, F-SAM (Li et al., 2024) improves generalization and convergence by employing an exponential moving average of historical stochastic gradients to eliminate the detrimental full-gradient component from SAM's perturbation, retaining only mini-batch gradient noise. Beyond algorithmic modifications, recent theoretical studies have provided deeper insights into the behavior of SAM and its variants (Andriushchenko & Flammarion, 2022; Si & Yun, 2024; Luo et al., 2024; Oikonomou & Loizou, 2025). In particular, Eigen-SAM (Luo et al., 2024) analyzes SAM through third-order stochastic differential equations, revealing that the critical role of alignment between the perturbation vector and the Hessian's principal eigenvector, and explicitly regularizes the top Hessian eigenvalue by enforcing such alignment.

Hessian Spectrum, Eigenvalues, and Generalization. Recent studies have highlighted the importance of the Hessian spectrum—the set of its eigenvalues—in characterizing model generalization, with the top Hessian eigenvalue plays a crucial role. Lyu et al. (Lyu et al., 2022) interpret the top Hessian eigenvalue as a direct measure of sharpness, showing that smaller values are typically associated with flatter minima. Cohen et al. (Cohen et al., 2021; Arora et al., 2022) further identify the Edge of Stability (EoS) phenomenon, in which the top eigenvalue hovers near the stability threshold during full-batch gradient descent training. Furthermore, Damian et al. (Damian et al., 2022) show that unstable deviations of gradient descent along the principal eigenvector induce a self-stabilizing effect that effectively constrains the top Hessian eigenvalue. Collectively, these results indicate that both the top Hessian eigenvalue and its corresponding principal eigenvector are crucial to characterizing sharpness and, consequently, the generalization behavior of models.

Theoretical Analysis and Convergence of SAM. A growing body of theoretical research has focused on understanding the optimization dynamics and convergence behavior of SAM (Wilson et al., 2017; Keskar et al., 2017; Soudry et al., 2018). Foret et al. (Foret et al., 2021) derive a PAC-Bayesian generalization bound, showing that SAM implicitly minimizes a local-entropy-based objective that favors flat minima. From an optimization perspective, Andriushchenko and Flammarion (Andriushchenko & Flammarion, 2022) establish a convergence rate of $O(1/\sqrt{T})$ for SAM under standard smooth, non-convex, and bounded-variance assumptions, while also theoretically characterizing its implicit bias and the generalization-enhancing mechanism using diagonal linear networks. Subsequently, Si and Yun (Si & Yun, 2024) show that under practical settings with a constant perturbation radius and gradient normalization, the convergence bound of SAM is affected by an unavoidable error term, implying convergence only to a neighborhood of the optimum. More recently, Oikonomou and Loizou (Oikonomou & Loizou, 2025) present a unified convergence analysis of SAM and its variants, deriving improved theoretical rates and clarifying how the interplay between step size, perturbation radius, and smoothness constants governs optimization stability. Beyond convergence guarantees, increasing attention has been paid to the role of explicit second-order information in improving generalization. In this vein, Luo et al. (Luo et al., 2024) propose Eigen-SAM, which aligns the perturbation direction with the principal Hessian eigenvector, enabling explicit reduction of the top Hessian eigenvalue and thereby enhancing generalization performance.

7. Conclusions

In this work, we conducted an in-depth analysis of gradient directions in SAM optimization. Motivated by the observation that the angle between the gradient and the principal eigenvector plays a critical role in guiding optimization toward flatter regions of the loss landscape, we propose the X-SAM method, a novel method that explicitly controls the top Hessian eigenvalue by enlarging this angle. We provide a theoretical convergence analysis for X-SAM, and extensive experimental results demonstrate its superiority over four state-of-the-art SAM-based methods, including SAM, WSAM, GSAM, and Eigen-SAM.

References

Ahn, K., Jadbabaie, A., and Sra, S. How to escape sharp minima with random perturbations. *arXiv preprint arXiv:2305.15659*, 2023.

Andriushchenko, M. and Flammarion, N. Towards understanding sharpness-aware minimization. *arXiv preprint arXiv:2206.06232*, 2022.

Arora, S., Li, Z., and Panigrahi, A. Understanding gradient descent on the edge of stability in deep learning. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*. PMLR, 2022.

Bahri, D., Mobahi, H., and Tay, Y. Sharpness-aware minimization improves language model generalization. *arXiv preprint arXiv:2110.08529*, 2021.

Barrett, D. G. and Dherin, B. Implicit gradient regularization. *arXiv preprint arXiv:2009.11162*, 2020.

Chaudhari, P., Choromanska, A., Soatto, S., LeCun, Y., Baldassi, C., Borgs, C., Chayes, J., and Sagun, L. Entropy-sgd: Biasing gradient descent into wide valleys. *Journal of Statistical Mechanics: Theory and Experiment*, (12): 124018, 2019.

Cohen, J. M., Yu, X., Barak, B., and Advani, M. S. Gradient descent on neural networks typically occurs at the edge of stability. *arXiv preprint arXiv:2103.00065*, 2021.

Damian, A., Nichani, E., and Lee, J. D. Self-stabilization: The implicit bias of gradient descent at the edge of stability. *arXiv preprint arXiv:2209.15594*, 2022.

Deng, L. The mnist database of handwritten digit images for machine learning research. *IEEE Signal Processing Magazine*, 29(6):141–142, 2012.

Drucker, H. and Cun, Y. L. Double backpropagation increasing generalization performance. In *IJCNN-91: Seattle International Joint Conference on Neural Networks*, volume 2, pp. 145–150. IEEE, 1991.

Foret, P., Kleiner, A., Mobahi, H., and Neyshabur, B. Sharpness-aware minimization for efficiently improving generalization. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.

He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.

Hochreiter, S. and Schmidhuber, J. Flat minima. *Neural Computation*, 9(1):1–42, 1997.

Jastrzębski, S., Kenton, Z. K., Arpit, D., Ballas, N., Fischer, A., Bengio, Y., and Storkey, A. On the relation between the sharpest directions of dnn loss and the sgd step length. In *International Conference on Learning Representations (ICLR) Workshop*, 2018.

Jastrzebski, S., Arpit, D., Astrand, O., Kerg, G. B., Wang, H., Xiong, C., Socher, R., Cho, K., and Geras, K. J. Catastrophic fisher explosion: Early phase fisher matrix impacts generalization. In *International Conference on Machine Learning (ICML)*, pp. 4772–4784. PMLR, 2021.

Jiang, Y., Neyshabur, B., Mobahi, H., Krishnan, D., and Bengio, S. Fantastic generalization measures and where to find them. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2019.

Kaddour, J., Liu, L., Silva, R., and Kusner, M. J. When do flat minima optimizers work? In *Advances in Neural Information Processing Systems*, volume 35, pp. 16577–16595, 2022.

Keskar, N. S., Mudigere, D., Nocedal, J., Smelyanskiy, M., and Tang, P. T. P. On large-batch training for deep learning: Generalization gap and sharp minima. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2017.

Kingma, D. P. and Ba, J. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015.

Krizhevsky, A., Sutskever, I., and Hinton, G. E. Imagenet classification with deep convolutional neural networks. *Communications of the ACM*, 60(6):84–90, 2017.

Li, T. et al. Friendly sharpness-aware minimization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

Liang, T., Poggio, T., Rakhlin, A., and Stokes, J. Fisher-rao metric, geometry, and complexity of neural networks. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 89 of *Proceedings of Machine Learning Research*, pp. 888–896. PMLR, 2019.

Liu, Y., Yu, S., and Lin, T. Regularizing deep neural networks with stochastic estimators of hessian trace. *arXiv preprint arXiv:2208.05924*, 2022.

Luo, H., Zhai, X., Xu, Y., and Sun, X. Explicit eigenvalue regularization improves sharpness-aware minimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 4424–4453, 2024.

Lyu, K., Li, Z., and Arora, S. Understanding the generalization benefit of normalization layers: Sharpness reduction.

In *Advances in Neural Information Processing Systems*, volume 35, pp. 34689–34708, 2022.

Netzer, Y., Wang, T., Coates, A., Bissacco, A., Wu, B., Ng, A. Y., et al. Reading digits in natural images with unsupervised feature learning. In *NIPS Workshop on Deep Learning and Unsupervised Feature Learning*, volume 2011, Granada, Spain, 2011.

Oikonomou, L. and Loizou, N. Sharpness-aware minimization: General analysis and improved rates. In *International Conference on Learning Representations (ICLR)*, 2025. Poster.

Si, D. and Yun, C. Practical sharpness-aware minimization cannot converge all the way to optima. In *Advances in Neural Information Processing Systems*, volume 36, 2024.

Song, M., Ahn, K., and Yun, C. Does sgd really happen in tiny subspaces? In *Advances in Neural Information Processing Systems*, volume 12, 2024.

Soudry, D., Hoffer, E., Nacson, M. S., Gunasekar, S., and Srebro, N. The implicit bias of gradient descent on separable data. *Journal of Machine Learning Research*, 19 (70):1–57, 2018.

Wilson, A. C., Roelofs, R., Stern, M., Srebro, N., and Recht, B. The marginal value of adaptive gradient methods in machine learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Xiao, H., Rasul, K., and Vollgraf, R. Fashion-mnist: A novel image dataset for benchmarking machine learning algorithms. In *arXiv preprint arXiv:1708.07747*, 2017.

Xie, Z., Sato, I., and Sugiyama, M. A diffusion theory for deep learning dynamics: Stochastic gradient descent exponentially favors flat minima. In *arXiv preprint arXiv:2002.03495*, 2020.

Yao, Z., Gholami, A., Keutzer, K., and Mahoney, M. W. Pyhessian: Neural networks through the lens of the hessian. In *2020 IEEE International Conference on Big Data (Big Data)*, pp. 581–590. IEEE, 2020.

Yue, Y., Wang, Z., and Zhang, Y. Sharpness-aware minimization revisited: Weighted sharpness as a regularization term. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023.

Zagoruyko, S. and Komodakis, N. Wide residual networks. *arXiv preprint arXiv:1605.07146*, 2016.

Zhuang, J., Gong, B., Yuan, L., Cui, Y., Adam, H., Dvornek, N. C., Tatikonda, S., Duncan, J. S., and Liu, T. Surrogate gap minimization improves sharpness-aware training. In *The Tenth International Conference on Learning Representations (ICLR)*, 2022.

A. Proofs of Theorem 4.1

Lemma A.1 (Principal eigenvector deviation under a spectral gap)**.** *We consider a symmetric matrix $H \in \mathbb{R}^{d\times d}$ with eigenvalues $\lambda_1(H) > \lambda_2(H) \geq \cdots$ and a spectral gap $\gamma := \lambda_1(H) - \lambda_2(H) > 0$. We define the perturbed matrix $H^+ := H + \Delta H$ where ΔH is symmetric, and we set the perturbation magnitude $s := \|\Delta H\|_{\mathrm{op}}$. We assume that the perturbation satisfies $s < \gamma$. We let $v_1 := v_1(H)$ and $v' := v_1(H^+)$ denote the unit principal eigenvectors of H and H^+, respectively, with the sign of v' chosen so that $v'^\top v_1 \geq 0$. Then we have*

$$\sin\angle(v', v_1) \leq \frac{s}{\gamma - s}, \qquad \|v' - v_1\| \leq \frac{2s}{\gamma - s}.$$

Proof. We let $u := v_1$ and $v := v'$. We decompose v into components parallel and orthogonal to u:

$$v = \alpha u + w, \qquad \alpha := u^\top v \geq 0, \qquad w \perp u, \qquad \|v\| = 1 \Rightarrow \alpha^2 + \|w\|^2 = 1.$$

We note that $\|w\| = \sin\angle(u, v)$.

We let $\mu_1 := \lambda_1(H^+)$ so that $(H + \Delta H)v = \mu_1 v$. Rearranging the terms, we obtain

$$(H - \mu_1 I)v = -\Delta H\, v.$$

We project onto $u^\perp$ using $P_\perp := I - uu^\top$. Since $P_\perp u = 0$ and $P_\perp v = w$, we arrive at

$$(H - \mu_1 I)w = -P_\perp \Delta H\, v.$$

Taking norms and using $\|P_\perp\| \leq 1$, we obtain

$$\|(H - \mu_1 I)w\| \leq \|\Delta H\|_{\mathrm{op}}\|v\| = s. \tag{A.1}$$

Next, we lower bound the left-hand side. By applying Weyl's inequality, we have $\mu_1 = \lambda_1(H^+) \geq \lambda_1(H) - \|\Delta H\|_{\mathrm{op}} = \lambda_1 - s$, so that

$$\mu_1 - \lambda_2 \geq (\lambda_1 - s) - \lambda_2 = \gamma - s. \tag{A.2}$$

Since $w \perp u$ lies in the invariant subspace corresponding to eigenvalues at most λ_2, we observe that the smallest eigenvalue of $(\mu_1 I - H)$ restricted to $u^\perp$ is at least $\mu_1 - \lambda_2$. Hence, we have

$$\|(\mu_1 I - H)w\| \geq (\mu_1 - \lambda_2)\|w\| \geq (\gamma - s)\|w\|. \tag{A.3}$$

Combining (A.1) and (A.3), we find $(\gamma - s)\|w\| \leq s$, i.e.,

$$\|w\| \leq \frac{s}{\gamma - s}.$$

Therefore, we conclude $\sin\angle(v', v_1) = \|w\| \leq s/(\gamma - s)$. Finally, because $v'^\top v_1 \geq 0$, we note that the angle $\theta := \angle(v', v_1) \in [0, \pi/2]$ and

$$\|v' - v_1\| = 2\sin(\theta/2) \leq 2\sin\theta = 2\|w\| \leq \frac{2s}{\gamma - s}.$$

This completes the proof. □

We consider the local geometry of a thrice-differentiable function $f : \mathbb{R}^d \to \mathbb{R}$. Let $H := \nabla^2 f(x)$ be the Hessian at x, with eigenvalues $\lambda_1(H) > \lambda_2(H) \geq \ldots$. Given an update step $\Delta x = \eta\tilde{h}$, the perturbed Hessian is $H^+ := \nabla^2 f(x + \Delta x)$, and we define the residual operator $\Delta H := H^+ - H$.

Definition A.2 (Spectral and Smoothness Parameters)**.** The following quantities characterize the stability of the system:

- **Eigengap:** $\gamma := \lambda_1(H) - \lambda_2(H) > 0$.
- **Perturbation Magnitude:** $s := \|\Delta H\|_{\mathrm{op}}$.

- **Uniform Third-derivative Bound:** A constant $M_3 > 0$ such that:

$$|\nabla^3 f(z)[a,b,c]| \le M_3 \|a\| \|b\| \|c\|, \quad \forall a,b,c \in \mathbb{R}^d.$$

Assumption 7 (Decrease Conditions)**.** *To guarantee the strict decrease of the top eigenvalue, we assume:*

- ***Non-degeneracy:*** *The perturbation is bounded by $s < \gamma$.*
- ***Descent Direction (C1):*** *$v_1^\top \Delta H v_1 \le -\eta\kappa$ for some $\kappa > 0$.*
- ***Step Size Sufficiency (C3):*** *$\eta\kappa > C_0 s$, where $C_0 := \frac{4(\|H\|_{\mathrm{op}}+s)}{\gamma - s}$ is the spectral sensitivity coefficient.*

Theorem A.3 (Strict Decrease of λ_1)**.** *Under Assumption 7, the perturbation magnitude satisfies $s \le M_3 \eta \|\tilde{h}\|$, and the leading eigenvalue of the perturbed Hessian satisfies:*

$$\lambda_1(H + \Delta H) \le \lambda_1(H) + v_1^\top \Delta H v_1 + \frac{4(\|H\|_{\mathrm{op}} + s)}{\gamma - s}\, s. \tag{13}$$

Furthermore, the conditions (C1) and (C3) imply $\lambda_1(H + \Delta H) < \lambda_1(H)$.

Proof. We apply the fundamental theorem of calculus to the matrix-valued function $t \mapsto \nabla^2 f(x + t\Delta x)$ to write

$$\Delta H = \int_0^1 \frac{d}{dt} \nabla^2 f(w + t\Delta x)\, dt = \int_0^1 \nabla^3 f(w + t\Delta w)[\,\cdot, \cdot, \Delta w\,]\, dt. \tag{14}$$

Consequently, for any unit vectors u, v, we bound the operator norm s as follows:

$$\begin{aligned} s := \|\Delta H\|_{\mathrm{op}} &= \sup_{\|u\|=\|v\|=1} \left| u^\top \Delta H\, v \right| \\ &= \sup_{\|u\|=\|v\|=1} \left| \int_0^1 \nabla^3 f(x + t\Delta x)[u, v, \Delta x]\, dt \right| \\ &\le \sup_{\|u\|=\|v\|=1} \int_0^1 M_3 \|u\|\, \|v\|\, \|\Delta x\|\, dt \\ &= M_3 \|\Delta x\| = M_3\, \eta\, \|\tilde{h}\|. \end{aligned} \tag{15}$$

Using the Rayleigh quotient, we express the leading eigenvalue as $\lambda_1(H + \Delta H) = {v'}^\top (H + \Delta H)\, v'$. By adding and subtracting $v_1^\top (H + \Delta H)\, v_1$, we decompose this term as

$$\lambda_1(H + \Delta H) = v_1^\top (H + \Delta H)\, v_1 + \Big({v'}^\top (H + \Delta H)\, v' - v_1^\top (H + \Delta H)\, v_1 \Big). \tag{16}$$

Moreover, utilizing the eigen-equation $H v_1 = \lambda_1(H) v_1$ and the normalization $\|v_1\| = 1$, we simplify the first term:

$$v_1^\top (H + \Delta H)\, v_1 = \lambda_1(H) + v_1^\top \Delta H\, v_1. \tag{17}$$

We recall that for any symmetric matrix A and unit vectors u, v, the inequality $\left| u^\top A u - v^\top A v \right| \le 2\|A\|_{\mathrm{op}} \|u - v\|$ holds. Applying this bound to our context, we obtain

$$\begin{aligned} \left| {v'}^\top (H + \Delta H) v' - v_1^\top (H + \Delta H) v_1 \right| &\le 2\|H + \Delta H\|_{\mathrm{op}} \|v' - v_1\| \\ &\le 2(\|H\|_{\mathrm{op}} + \|\Delta H\|_{\mathrm{op}}) \|v' - v_1\| \\ &= 2(\|H\|_{\mathrm{op}} + s) \|v' - v_1\|. \end{aligned} \tag{18}$$

To estimate the eigenvector difference, we set $u := v_1$ and $v := v'$. We decompose v as

$$v = \alpha u + w, \qquad \alpha = u^\top v \ge 0, \qquad w \perp u, \qquad \alpha^2 + \|w\|^2 = 1.$$

We define $\mu_1 := \lambda_1(H + \Delta H)$. Since $(H + \Delta H)v = \mu_1 v$, we rearrange the equation to

$$(H - \mu_1 I)v = -\Delta H\, v. \tag{19}$$

We introduce $P^\perp := I - uu^\top$ as the orthogonal projector onto $u^\perp$. Applying $P^\perp$ to (19) and utilizing the properties $P^\perp u = 0$ and $P^\perp v = w$, we derive

$$(H - \mu_1 I)w = -P^\perp \Delta H\, v. \tag{20}$$

Taking the norms, we obtain

$$\|(H - \mu_1 I)w\| \le \|P^\perp\|_{\mathrm{op}} \|\Delta H\|_{\mathrm{op}} \|v\| \le s, \tag{21}$$

where we used $\|P^\perp\|_{\mathrm{op}} = 1$ and $\|v\| = 1$. Next, by invoking Weyl's inequality, we have

$$\mu_1 = \lambda_1(H + \Delta H) \ge \lambda_1(H) - \|\Delta H\|_{\mathrm{op}} = \lambda_1(H) - s, \tag{22}$$

and hence

$$\mu_1 - \lambda_2(H) \ge (\lambda_1(H) - s) - \lambda_2(H) = \gamma - s. \tag{23}$$

Since $w \perp u$, the vector w lies in the orthogonal complement of the top eigenspace of H. Therefore, we lower bound the norm as

$$\|(\mu_1 I - H)w\| \ge (\mu_1 - \lambda_2(H))\|w\| \ge (\gamma - s)\|w\|. \tag{24}$$

We note that $\|(H - \mu_1 I)w\| = \|(\mu_1 I - H)w\|$. Combining (21) and (24), we find that

$$(\gamma - s)\|w\| \le s \quad \Rightarrow \quad \|w\| \le \frac{s}{\gamma - s}.$$

Finally, since $\alpha \ge 0$ (implying the angle between u and v lies in $[0, \pi/2]$), we have $\|v - u\| \le 2\|w\|$, and thus we conclude

$$\|v' - v_1\| = \|v - u\| \le \frac{2s}{\gamma - s}. \tag{25}$$

Substituting (17) into (16) and utilizing the bounds (18) and (25), we obtain

$$\lambda_1(H + \Delta H) \le \lambda_1(H) + v_1^\top \Delta H v_1 + 2(\|H\|_{\mathrm{op}} + s) \cdot \frac{2s}{\gamma - s}.$$

This can be rewritten as

$$\lambda_1(H + \Delta H) \le \lambda_1(H) + v_1^\top \Delta H v_1 + \underbrace{\frac{4(\|H\|_{\mathrm{op}} + s)}{\gamma - s}}_{=:C_0} s. \tag{26}$$

If condition (C1) holds, i.e., $v_1^\top \Delta H v_1 \le -\eta\kappa$, then (26) implies $\lambda_1(H + \Delta H) \le \lambda_1(H) - \eta\kappa + C_0 s$. If in addition (C3) is satisfied, i.e., $\eta\kappa > C_0 s$, then we strictly obtain

$$\lambda_1(H + \Delta H) < \lambda_1(H), \tag{27}$$

which proves the desired strict decrease. □

B. Proofs of Theorem 4.2

Definition B.1 (Algorithm Quantities)**.** We define the following quantities to characterize the convergence rate and perturbation bias:

- **Gradient Notations:** Let $g_t := \nabla f(w_t)$ be the true gradient, $h_t := \nabla f(w_t + \rho\epsilon_t)$ be the perturbed gradient, and $\hat{h}_t$ be the stochastic estimate.
- **SAM Bias:** $D_\rho := \beta^2\rho^2(1 + C_1\alpha^2)$, representing the error upper bound $\mathbb{E}\|g_t - h_t\|^2 \le D_\rho$.
- **Descent Coefficient:** $c_0 := \frac{3}{4} - 2\alpha^2$, which serves as the effective curvature scaling.

Assumption 8 (Stochastic Setup and Parameters)**.** *We adopt the following standard assumptions:*

1. ***Oracle Properties:*** *The stochastic gradient is unbiased,* $\mathbb{E}[\hat{h}_t|w_t] = h_t$*, with bounded variance* $\mathbb{E}[\|\hat{h}_t - h_t\|^2] \le \sigma^2$*.*
2. ***Perturbation Bound:*** *The noise satisfies* $\|\epsilon_t\|^2 \le 1 + C_1\alpha^2$*.*
3. ***Parameter Selection:*** *We choose* α *such that* $\alpha^2 < 3/8$ *(ensuring* $c_0 > 0$*) and set the step size as:*

$$\eta = \min\left\{\frac{1}{2\beta}, \sqrt{\frac{\Delta}{\beta\sigma^2 T}}\right\}, \quad \textit{where } \Delta := f(w_0) - f_*.$$

Theorem B.2 (Convergence Rate)**.** *Under Assumption 8, the X-SAM algorithm guarantees the following bound on the average squared gradient norm:*

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \;\le\; c_1\left(\frac{\beta\Delta}{T} + \sqrt{\frac{\beta\Delta\sigma^2}{T}}\right) + c_2(1+2\alpha^2)D_\rho, \tag{28}$$

where $c_1, c_2 > 0$ *are constants depending only on the descent coefficient* c_0*.*

Proof. We assume $\alpha \in [0, 2]$ and $\|v_t\| = 1$. Consequently, for the projector $P_t := I - \alpha v_t v_t^\top$, we have the operator norm bound $\|P_t\|_{\text{op}} = \max\{1, |1-\alpha|\} \le 1$. Let us define the gradient terms: $\|P_t\|_{\text{op}} = \max\{1, |1-\alpha|\} \le 1$. Let

$$g_t := \nabla f(w_t), \qquad h_t := \nabla f(w_t + \rho\epsilon_t), \qquad \hat{h}_t := \nabla f(w_t + \rho\epsilon_t; \xi_t), \qquad \tilde{h}_t := P_t\hat{h}_t,$$

and the update rule $w_{t+1} = w_t - \eta\tilde{h}_t$. We assume that the stochastic gradient is unbiased with bounded variance, i.e.,

$$\mathbb{E}[\hat{h}_t \mid w_t, \epsilon_t, v_t] = h_t, \qquad \mathbb{E}\Big[\|\hat{h}_t - h_t\|^2 \mid w_t, \epsilon_t, v_t\Big] \le \sigma^2.$$

By the tower property, this implies the global statements

$$\mathbb{E}\langle u_t, \hat{h}_t - h_t\rangle = 0 \;\; (\forall\, u_t = u_t(w_t, \epsilon_t, v_t)), \qquad \mathbb{E}\|\hat{h}_t - h_t\|^2 \le \sigma^2. \tag{29}$$

Assume $\|\epsilon_t\|^2 \le 1 + C_1\alpha^2$ for some $C_1 > 0$. Since ∇f is β-Lipschitz, we have the pointwise bound

$$\|g_t - h_t\| = \big\|\nabla f(w_t) - \nabla f(w_t + \rho\epsilon_t)\big\| \le \beta\rho\|\epsilon_t\| = \beta\rho \tag{30}$$

By the β-smoothness of f, we have

$$f(w_{t+1}) \le f(w_t) + \langle g_t, w_{t+1} - w_t\rangle + \frac{\beta}{2}\|w_{t+1} - w_t\|^2 = f(w_t) - \eta\langle g_t, \tilde{h}_t\rangle + \frac{\beta\eta^2}{2}\|\tilde{h}_t\|^2.$$

Taking the total expectation yields

$$\mathbb{E}f(w_{t+1}) \le \mathbb{E}f(w_t) - \eta\,\mathbb{E}\langle g_t, \tilde{h}_t\rangle + \frac{\beta\eta^2}{2}\mathbb{E}\|\tilde{h}_t\|^2. \tag{31}$$

Bounding the inner product term: Since $\tilde{h}_t = P_t\hat{h}_t$, using the tower property and the definition of P_t, we derive:

$$\begin{aligned}
\mathbb{E}\langle g_t, \tilde{h}_t\rangle &= \mathbb{E}\big[\langle g_t, P_t\hat{h}_t\rangle\big] = \mathbb{E}\Big[\mathbb{E}\big[\langle g_t, P_t\hat{h}_t\rangle \mid w_t, \epsilon_t, v_t\big]\Big] \\
&= \mathbb{E}\Big[\big\langle g_t, P_t\,\mathbb{E}[\hat{h}_t \mid w_t, \epsilon_t, v_t]\big\rangle\Big] = \mathbb{E}\langle g_t, P_t h_t\rangle \\
&= \mathbb{E}\langle g_t, h_t\rangle - \alpha\,\mathbb{E}[\langle h_t, v_t\rangle\langle g_t, v_t\rangle]\,.
\end{aligned} \tag{32}$$

Since $\|P_t\|_{\text{op}} \le 1$, we have $\|\tilde{h}_t\| = \|P_t\hat{h}_t\| \le \|\hat{h}_t\|$ pointwise. Consequently, this yields the inequality:

$$\mathbb{E}\|\tilde{h}_t\|^2 \le \mathbb{E}\|\hat{h}_t\|^2. \tag{33}$$

We decompose the stochastic gradient as $\hat{h}_t = h_t + (\hat{h}_t - h_t)$ and apply (29) to derive:

$$\begin{aligned}
\mathbb{E}\|\hat{h}_t\|^2 &= \mathbb{E}\langle \hat{h}_t, \hat{h}_t\rangle \\
&= \mathbb{E}\Big\langle h_t + (\hat{h}_t - h_t),\, h_t + (\hat{h}_t - h_t)\Big\rangle \\
&= \mathbb{E}\Big[\|h_t\|^2 + 2\langle h_t, \hat{h}_t - h_t\rangle + \|\hat{h}_t - h_t\|^2\Big] \\
&= \mathbb{E}\|h_t\|^2 + 2\,\mathbb{E}\langle h_t, \hat{h}_t - h_t\rangle + \mathbb{E}\|\hat{h}_t - h_t\|^2 \\
&= \mathbb{E}\|h_t\|^2 + \mathbb{E}\|\hat{h}_t - h_t\|^2 \\
&\le \mathbb{E}\|h_t\|^2 + \sigma^2. \qquad (34)
\end{aligned}$$

Combining (33) and (34), we deduce that

$$\mathbb{E}\|\tilde{h}_t\|^2 \le \mathbb{E}\|h_t\|^2 + \sigma^2. \qquad (35)$$

Substituting (32) and (35) into (31), we arrive at

$$\mathbb{E}f(w_{t+1}) \le \mathbb{E}f(w_t) - \eta\,\mathbb{E}\langle g_t, h_t\rangle + \eta\alpha\,\mathbb{E}[\langle h_t, v_t\rangle\langle g_t, v_t\rangle] + \frac{\beta\eta^2}{2}\mathbb{E}\|h_t\|^2 + \frac{\beta\eta^2}{2}\sigma^2. \qquad (36)$$

We utilize the polarization identity $2\langle a, b\rangle = \|a\|^2 + \|b\|^2 - \|a-b\|^2$ (pointwise) and take the expectation to write

$$-\eta\,\mathbb{E}\langle g_t, h_t\rangle = -\frac{\eta}{2}\mathbb{E}\|g_t\|^2 - \frac{\eta}{2}\mathbb{E}\|h_t\|^2 + \frac{\eta}{2}\mathbb{E}\|g_t - h_t\|^2. \qquad (37)$$

For any $\delta \in (0,1)$, we apply Young's inequality using the fact that $|v_t| = 1$ to obtain the pointwise bound:

$$\begin{aligned}
\alpha\langle h_t, v_t\rangle\langle g_t, v_t\rangle &= \Big(\alpha\langle h_t, v_t\rangle\Big)\Big(\langle g_t, v_t\rangle\Big) \\
&\le \frac{\delta}{2}\,\langle g_t, v_t\rangle^2 + \frac{1}{2\delta}\Big(\alpha\langle h_t, v_t\rangle\Big)^2 \qquad (\forall\,\delta > 0) \qquad (38) \\
&= \frac{\delta}{2}\,\langle g_t, v_t\rangle^2 + \frac{\alpha^2}{2\delta}\,\langle h_t, v_t\rangle^2 \\
&\le \frac{\delta}{2}\,\|g_t\|^2 + \frac{\alpha^2}{2\delta}\,\|h_t\|^2, \qquad (39)
\end{aligned}$$

Hence, we have

$$\eta\alpha\,\mathbb{E}[\langle h_t, v_t\rangle\langle g_t, v_t\rangle] \le \frac{\eta\delta}{2}\mathbb{E}\|g_t\|^2 + \frac{\eta\alpha^2}{2\delta}\mathbb{E}\|h_t\|^2. \qquad (40)$$

Substituting (37) and (40) into (36), we obtain

$$\begin{aligned}
\mathbb{E}f(w_{t+1}) \le\ & \mathbb{E}f(w_t) - \frac{\eta}{2}(1-\delta)\mathbb{E}\|g_t\|^2 + \frac{\eta}{2}\mathbb{E}\|g_t - h_t\|^2 \\
&+ \Big(-\frac{\eta}{2} + \frac{\eta\alpha^2}{2\delta} + \frac{\beta\eta^2}{2}\Big)\mathbb{E}\|h_t\|^2 + \frac{\beta\eta^2}{2}\sigma^2. \qquad (41)
\end{aligned}$$

From the triangle inequality $\|h_t\| \le \|g_t\| + \|h_t - g_t\|$, we have

$$\|h_t\|^2 \le 2\|g_t\|^2 + 2\|h_t - g_t\|^2. \qquad (42)$$

Taking the expectation and using (30) (which implies $\|h_t - g_t\|^2 = \|g_t - h_t\|^2 \le D_\rho$ point-wise), we obtain

$$\mathbb{E}\|h_t\|^2 \le 2\,\mathbb{E}\|g_t\|^2 + 2D_\rho, \qquad \mathbb{E}\|g_t - h_t\|^2 \le D_\rho. \qquad (43)$$

We fix $\delta = \frac{1}{2}$, $\beta\eta \le \frac{1}{2}$, and $\alpha \le \sqrt{\frac{3}{8}}$, so that we satisfy the descent condition

$$\frac{\alpha^2}{\delta} + \beta\eta \le \frac{3-\delta}{2} \qquad (44)$$

Applying (43), we obtain

$$\begin{aligned}\mathbb{E}f(w_{t+1}) &\le \mathbb{E}f(w_t) - \frac{\eta}{2}(1-\delta)\mathbb{E}\|g_t\|^2 + \frac{\eta}{2}D_\rho + \Big(-\frac{\eta}{2} + \frac{\eta\alpha^2}{2\delta} + \frac{\beta\eta^2}{2}\Big)\Big(2\,\mathbb{E}\|g_t\|^2 + 2D_\rho\Big) + \frac{\beta\eta^2}{2}\sigma^2 \\ &\le \mathbb{E}f(w_t) - \eta\Big(\frac{3-\delta}{2} - \frac{\alpha^2}{\delta} - \beta\eta\Big)\mathbb{E}\|g_t\|^2 + \eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2}{2}\sigma^2 \qquad (45)\end{aligned}$$

We define

$$c := \frac{3-\delta}{2} - \frac{\alpha^2}{\delta} - \beta\eta.$$

We fix $\delta = \frac{1}{2}$ and assume $\beta\eta \le \frac{1}{2}$ and $\alpha^2 < \frac{3}{8}$. Then for this c we have

$$c = \frac{5}{4} - 2\alpha^2 - \beta\eta \ge \frac{5}{4} - 2\alpha^2 - \frac{1}{2} = \frac{3}{4} - 2\alpha^2 =: c_0 > 0,$$

which implies $1/c \le 1/c_0$.

From (45) we directly obtain

$$\eta c\,\mathbb{E}\|g_t\|^2 \le \mathbb{E}f(w_t) - \mathbb{E}f(w_{t+1}) + \eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2}{2}\sigma^2, \qquad (46)$$

$$\begin{aligned}\eta c\sum_{t=0}^{T-1}\mathbb{E}\|g_t\|^2 &\le \sum_{t=0}^{T-1}\big(\mathbb{E}f(w_t) - \mathbb{E}f(w_{t+1})\big) + T\eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2 T}{2}\sigma^2 \\ &= \mathbb{E}\big[f(w_0) - f(w_T)\big] + T\eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2 T}{2}\sigma^2 \\ &\le \Delta + T\eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2 T}{2}\sigma^2,\end{aligned}$$

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{\Delta}{c\,\eta T} + \frac{1}{c}\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta}{2c}\sigma^2 \qquad (47)$$

$$\le \frac{1}{c_0}\cdot\frac{\Delta}{\eta T} + \frac{1}{c_0}\Big(\frac{1}{2} + 2\alpha^2 + \beta\eta\Big)D_\rho + \frac{1}{c_0}\cdot\frac{\beta\eta}{2}\sigma^2. \qquad (48)$$

Using $\beta\eta \le \frac{1}{2}$, we further obtain

$$\Big(\frac{1}{2} + 2\alpha^2 + \beta\eta\Big)D_\rho \le (1 + 2\alpha^2)\,D_\rho.$$

Moreover, by the definition of η, we have

$$\frac{\Delta}{\eta T} \le 2\frac{\beta\Delta}{T} + \sqrt{\frac{\beta\Delta\sigma^2}{T}}, \qquad \beta\eta\sigma^2 \le \sqrt{\frac{\beta\Delta\sigma^2}{T}}.$$

We choose

$$\eta = \min\left\{\frac{1}{2\beta},\ \sqrt{\frac{\Delta}{\beta\sigma^2 T}}\right\}.$$

Plugging these bounds into (48), we obtain that there exist constants $c_1, c_2 > 0$ (depending only on c_0) such that

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le c_1\left(\frac{\beta\Delta}{T} + \sqrt{\frac{\beta\Delta\sigma^2}{T}}\right) + c_2(1 + 2\alpha^2)\,D_\rho, \qquad D_\rho = \beta^2\rho^2. \qquad (49)$$

In particular, we note that when $\sigma^2 > 0$ the stochastic term scales as $O(1/\sqrt{T})$, and when $\sigma = 0$ the bound reduces to $O(1/T)$ up to the perturbation bias term proportional to D_ρ.

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{\Delta}{c\,\eta T} + \frac{1}{c}\Big(\frac{1}{2}+\frac{\alpha^2}{\delta}+\beta\eta\Big)D_\rho + \frac{\beta\eta}{2c}\sigma^2 \tag{50}$$

$$\le \frac{1}{c_0}\cdot\frac{\Delta}{\eta T} + \frac{1}{c_0}\Big(\frac{1}{2}+\frac{\alpha^2}{\delta}+\beta\eta\Big)D_\rho + \frac{1}{c_0}\cdot\frac{\beta\eta}{2}\sigma^2$$

$$\le \frac{1}{c_0}\cdot\frac{\Delta}{\eta T} + \frac{1}{c_0}\Big(\frac{1}{2}+2\alpha^2+\beta\eta\Big)D_\rho + \frac{1}{c_0}\cdot\frac{\beta\eta}{2}\sigma^2. \tag{51}$$

Using $\beta\eta \le \frac{1}{2}$, we further obtain

$$\Big(\tfrac{1}{2}+2\alpha^2+\beta\eta\Big)D_\rho \le (1+2\alpha^2)\,D_\rho. \tag{52}$$

Substituting (52) into (48), we obtain

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{1}{c_0}\cdot\frac{\Delta}{\eta T} + \frac{1}{c_0}(1+2\alpha^2)\,D_\rho + \frac{1}{c_0}\cdot\frac{\beta\eta}{2}\sigma^2. \tag{53}$$

We choose the step size $\eta = \min\left\{\frac{1}{2\beta},\ \sqrt{\frac{\Delta}{\beta\sigma^2 T}}\right\}$. Then $\beta\eta \le \frac{1}{2}$ and, moreover, $\frac{\Delta}{\eta T} \le 2\frac{\beta\Delta}{T} + \sqrt{\frac{\beta\Delta\sigma^2}{T}}, \beta\eta\sigma^2 \le \sqrt{\frac{\beta\Delta\sigma^2}{T}}$. We obtain

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{1}{c_0}\left(2\frac{\beta\Delta}{T}+\sqrt{\frac{\beta\Delta\sigma^2}{T}}\right) + \frac{1}{c_0}(1+2\alpha^2)\,D_\rho + \frac{1}{2c_0}\sqrt{\frac{\beta\Delta\sigma^2}{T}}$$

$$= \frac{2}{c_0}\frac{\beta\Delta}{T} + \frac{3}{2c_0}\sqrt{\frac{\beta\Delta\sigma^2}{T}} + \frac{1}{c_0}(1+2\alpha^2)\,D_\rho \tag{54}$$

$$\le c_1\left(\frac{\beta\Delta}{T}+\sqrt{\frac{\beta\Delta\sigma^2}{T}}\right) + c_2(1+2\alpha^2)\,D_\rho.$$

□

Theorem B.3 (First-order change of the leading eigenvalue)**.** *We assume that $f:\mathbb{R}^d\to\mathbb{R}$ is $\mathcal{C}^4$ in a neighborhood of w_t, and we define $H(w)=\nabla^2 f(w)$ and $H_t := H(w_t)$. We assume the existence of $\rho>0$ such that the third and fourth derivatives are locally bounded: $M_3 := \sup_{\|u-w_t\|\le\rho}\|\nabla^3 f(u)\|_{\mathrm{op}} < \infty$, $M_4 := \sup_{\|u-w_t\|\le\rho}\|\nabla^4 f(u)\|_{\mathrm{op}} < \infty$. We further assume that H_t possesses a simple largest eigenvalue $\lambda_1(w_t)$ with unit eigenvector $v_1(w_t)$, and we define the spectral gap as $\gamma := \lambda_1(w_t)-\lambda_2(w_t) > 0$. We consider the update $\Delta w := w_{t+1}-w_t = -\eta g_t$ with $\eta>0$ chosen sufficiently small to ensure that $\|H(w_t+\Delta w)-H(w_t)\|_{\mathrm{op}} < \gamma/2$. We additionally assume that $\|(I-v_1v_1^\top)\nabla\lambda_1(w_t)\| \le \varepsilon$, and$v_1(w_t)^\top g_t = 0$. Under these assumptions, we establish that the leading eigenvalue increment $\Delta\lambda_1 := \lambda_1(w_{t+1})-\lambda_1(w_t)$ satisfies*

$$|\Delta\lambda_1| \le \eta\varepsilon\|g_t\| \;+\; O\big(\eta^2\|g_t\|^2\big) \;+\; O\Big(\frac{\eta^2\|g_t\|^2}{\gamma}\Big),$$

where we note that the hidden constants depend only on M_3, M_4 and on the universal spectral constant in the perturbation remainder.

Proof. We let $H(w)=\nabla^2 f(w)$ and $H_t := H(w_t)$. We define the updates as

$$\Delta w := w_{t+1}-w_t = -\eta g_t, \qquad \Delta H := H(w_t+\Delta w)-H(w_t).$$

By invoking the *analytic perturbation theorem* (Kato, 1966), we assert that there exists a neighborhood of H_t (in particular, whenever $\|\Delta H\|_{\mathrm{op}} < \gamma/2$) on which $\lambda_1(\cdot)$ is differentiable and

$$D\lambda_1(H_t)[E] = v_1^\top E\, v_1.$$

Hence, the first-order expansion yields

$$\Delta\lambda_1 := \lambda_1(w_{t+1}) - \lambda_1(w_t) = v_1^\top(\Delta H)\, v_1 + r_{\text{spec}}, \tag{55}$$

with the gap-controlled remainder satisfying

$$|r_{\text{spec}}| \le C_{\text{spec}} \frac{\|\Delta H\|_{\text{op}}^2}{\gamma}. \tag{56}$$

We let $\mathcal{B}$ be a ball around w_t containing the segment $\{w_t + s\Delta w : s \in [0,1]\}$, and we define the Lipschitz constants

$$L_0 := \sup_{w\in\mathcal{B}} \|DH(w)\|_{\text{op}}, \qquad L_1 := \sup_{w\in\mathcal{B}} \|D^2H(w)\|_{\text{op}}.$$

(In particular, if $\|\nabla^3 f(w)\|$ is bounded on $\mathcal{B}$ by M_3, then $L_0 \le M_3$. If moreover $\|\nabla^4 f(w)\|$ is bounded on $\mathcal{B}$ by M_4, then $L_1 \le M_4$.) Consequently, we bound the perturbation norm as

$$\|\Delta H\|_{\text{op}} = \|H(w_t + \Delta w) - H(w_t)\|_{\text{op}} \le L_0\|\Delta w\|. \tag{57}$$

We define $\phi(s) := H(w_t + s\Delta w)$. Then, noting that $\phi'(s) = DH(w_t + s\Delta w)[\Delta w]$, we write

$$\Delta H = \int_0^1 DH(w_t + s\Delta w)[\Delta w]\, ds. \tag{58}$$

Adding and subtracting $DH(w_t)[\Delta w]$, we obtain

$$\Delta H = DH(w_t)[\Delta w] + R_H, \qquad R_H := \int_0^1 \big(DH(w_t + s\Delta w) - DH(w_t)\big)[\Delta w]\, ds. \tag{59}$$

In coordinates, we express the differential as

$$DH(w_t)[\Delta w] = \sum_{i=1}^{d} \frac{\partial H}{\partial w_i}(w_t)\, \Delta w_i =: \nabla_w H[w_t] \cdot \Delta w. \tag{60}$$

Since D^2H is bounded by L_1 on $\mathcal{B}$, the map DH is Lipschitz on $\mathcal{B}$ with constant L_1, i.e.,

$$\|DH(u) - DH(w)\|_{\text{op}} \le L_1\|u - w\| \qquad \forall\, u, w \in \mathcal{B}.$$

Therefore, for all $s \in [0,1]$, we have

$$\|DH(w_t + s\Delta w) - DH(w_t)\|_{\text{op}} \le L_1\|s\Delta w\| = sL_1\|\Delta w\|.$$

Substituting this bound into the definition of R_H, we derive

$$\begin{aligned}
\|R_H\|_{\text{op}} &\le \int_0^1 \Big\| \big(DH(w_t + s\Delta w) - DH(w_t)\big)[\Delta w] \Big\|_{\text{op}}\, ds \\
&\le \int_0^1 \|DH(w_t + s\Delta w) - DH(w_t)\|_{\text{op}}\, \|\Delta w\|\, ds \\
&\le \int_0^1 sL_1\|\Delta w\|^2\, ds = \tfrac{1}{2}L_1\|\Delta w\|^2.
\end{aligned} \tag{61}$$

Combining (55) and (59), we arrive at

$$\Delta\lambda_1 = v_1^\top\big(DH(w_t)[\Delta w]\big)v_1 \;+\; v_1^\top R_H v_1 \;+\; r_{\text{spec}}. \tag{62}$$

Using $\|v_1\| = 1$ and (61), we find

$$|v_1^\top R_H v_1| \le \|R_H\|_{\text{op}} \le \tfrac{1}{2}L_1\|\Delta w\|^2.$$

Moreover, by applying (56) and (57), we obtain

$$|r_{\text{spec}}| \le C_{\text{spec}} \frac{\|\Delta H\|_{\text{op}}^2}{\gamma} \le C_{\text{spec}} \frac{L_0^2}{\gamma} \|\Delta w\|^2.$$

Expanding $DH(w_t)[\Delta w]$ using (60) gives

$$v_1^\top \big(DH(w_t)[\Delta w]\big) v_1 = \sum_{i=1}^{d} \Delta w_i \, v_1^\top \frac{\partial H}{\partial w_i}(w_t) v_1.$$

We let $H_t := H(w_t)$ and suppose the top eigenpair (λ_1, v_1) satisfies $H_t v_1 = \lambda_1 v_1, \qquad \|v_1\|_2 = 1$. We differentiate both sides with respect to the coordinate w_i (evaluated at w_t). Using the product rule, we obtain

$$\frac{\partial H}{\partial w_i}(w_t)\, v_1 \;+\; H_t \frac{\partial v_1}{\partial w_i} = \frac{\partial \lambda_1}{\partial w_i}(w_t)\, v_1 \;+\; \lambda_1 \frac{\partial v_1}{\partial w_i}. \tag{63}$$

Multiplying by $v_1^\top$ from the left, we obtain

$$v_1^\top \frac{\partial H}{\partial w_i}(w_t)\, v_1 \;+\; v_1^\top H_t \frac{\partial v_1}{\partial w_i} = \frac{\partial \lambda_1}{\partial w_i}(w_t)\, v_1^\top v_1 \;+\; \lambda_1\, v_1^\top \frac{\partial v_1}{\partial w_i}. \tag{64}$$

Since H_t is symmetric, we note that

$$v_1^\top H_t \frac{\partial v_1}{\partial w_i} = \lambda_1\, v_1^\top \frac{\partial v_1}{\partial w_i}, \tag{65}$$

so these terms cancel on both sides. Therefore, we derive

$$\frac{\partial \lambda_1}{\partial w_i}(w_t) = v_1^\top \frac{\partial H}{\partial w_i}(w_t)\, v_1, \qquad i = 1, \ldots, d. \tag{66}$$

and hence

$$v_1^\top \big(DH(w_t)[\Delta w]\big) v_1 = \sum_{i=1}^{d} \Delta w_i \frac{\partial \lambda_1}{\partial w_i}(w_t) = \langle \nabla \lambda_1(w_t),\, \Delta w \rangle. \tag{67}$$

Substituting (67) into (62) and using $\Delta w = -\eta g_t$, we obtain

$$\Delta \lambda_1 = -\eta \langle \nabla \lambda_1(w_t),\, g_t \rangle \;+\; v_1^\top R_H v_1 \;+\; r_{\text{spec}}.$$

Using the bounds above and the relation $\|\Delta w\| = \eta \|g_t\|$, we have

$$\Delta \lambda_1 = -\eta \langle \nabla \lambda_1(w_t),\, g_t \rangle \;+\; O\big(\eta^2 \|g_t\|^2\big) \;+\; O\Big(\frac{\eta^2 \|g_t\|^2}{\gamma}\Big),$$

where the hidden constants depend only on the local smoothness constants L_0, L_1 and on C_{spec}.

We decompose the gradient into the component orthogonal to v_1 and the component parallel to v_1:

$$\nabla \lambda_1(w_t) = (I - v_1 v_1^\top) \nabla \lambda_1(w_t) + v_1 v_1^\top \nabla \lambda_1(w_t). \tag{68}$$

Taking the inner product with Δw, we obtain

$$\langle \nabla \lambda_1(w_t), \Delta w \rangle = \langle (I - v_1 v_1^\top) \nabla \lambda_1(w_t), \Delta w \rangle + \langle v_1 v_1^\top \nabla \lambda_1(w_t), \Delta w \rangle \tag{69}$$

By applying the Cauchy–Schwarz inequality, we have

$$\big|\langle (I - v_1 v_1^\top) \nabla \lambda_1(w_t), \Delta w \rangle\big| \le \|(I - v_1 v_1^\top) \nabla \lambda_1(w_t)\| \, \|\Delta w\| \le \varepsilon \|\Delta w\|. \tag{70}$$

Moreover, using the identity $v_1 v_1^\top x = (v_1^\top x)\, v_1$ for any vector x, we observe that

$$v_1 v_1^\top \nabla \lambda_1(w_t) = \big(v_1^\top \nabla \lambda_1(w_t)\big)\, v_1, \tag{71}$$

$$\big\langle v_1 v_1^\top \nabla \lambda_1(w_t), \Delta w \big\rangle = \big(v_1^\top \nabla \lambda_1(w_t)\big)\, \big(v_1^\top \Delta w\big), \tag{72}$$

$$\big|\big\langle v_1 v_1^\top \nabla \lambda_1(w_t), \Delta w \big\rangle\big| \le \big|v_1^\top \nabla \lambda_1(w_t)\big| \, \big|v_1^\top \Delta w\big|. \tag{73}$$

Therefore, in the orthogonal case where $v_1^\top g_t = 0$, we deduce

$$\big|\eta\langle \nabla\lambda_1(w_t), g_t\rangle\big| = \big|\langle \nabla\lambda_1(w_t), \Delta w\rangle\big| \le \varepsilon\|\Delta w\| = \eta\varepsilon\|g_t\|.$$

Finally, we obtain the single-step eigenvalue change bound

$$|\Delta\lambda_1| \le \eta\varepsilon\|g_t\| \;+\; O\big(\eta^2\|g_t\|^2\big) \;+\; O\Big(\frac{\eta^2\|g_t\|^2}{\gamma}\Big),$$

where we note that the hidden constants depend only on the local smoothness constants L_0, L_1 and on C_{spec}. Hence, we conclude that when ε is sufficiently small, the first-order contribution is suppressed and λ_1 changes only by higher-order terms in one step. □

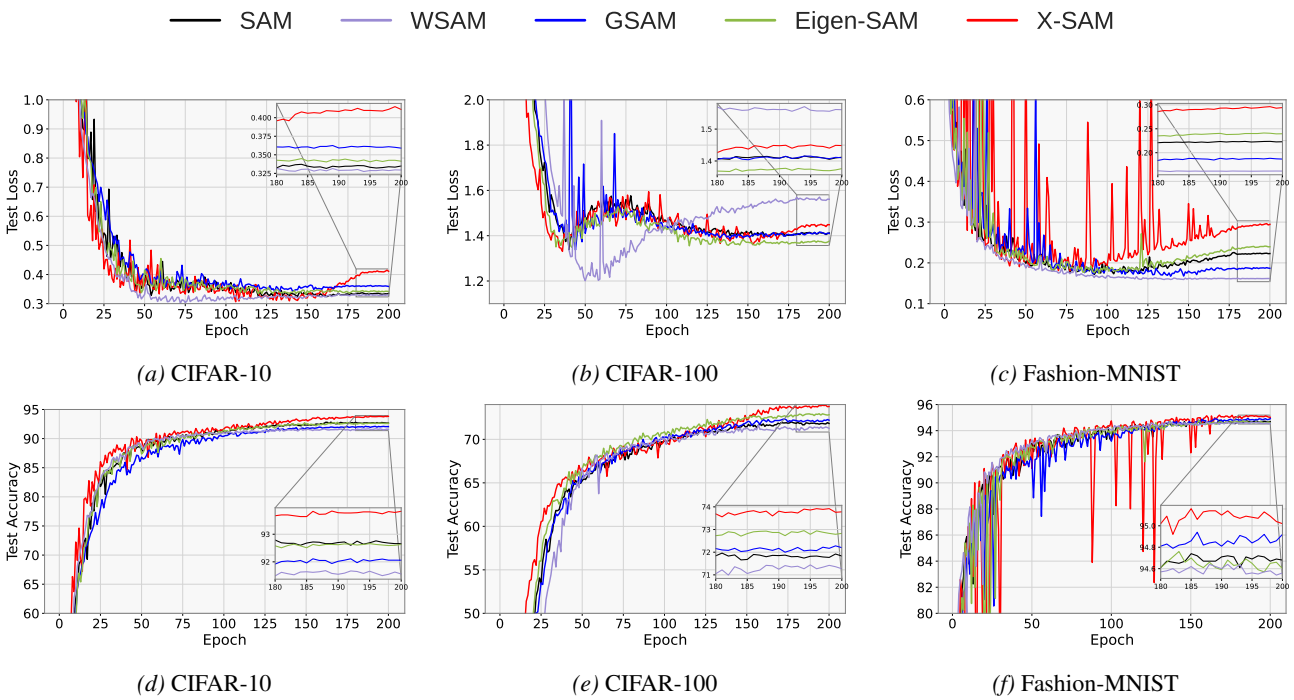
C. Additional Experimental Results

In this appendix, we present comprehensive empirical evaluations to further validate the effectiveness and robustness of the proposed X-SAM. First, we detail the evolution of test loss and accuracy across various architectures (refer to Section C.1), providing insights into the decoupling of loss minimization and generalization improvement. Then, we conduct sensitivity analyses on key hyperparameters, specifically investigating the impact of the perturbation radius ρ (explicated in Section C.2) and the gradient correction coefficient α (analyzed in Section C.3). Lastly, we perform a Hessian spectrum investigation (outlined in Section C.4) to visualize the eigenvalue distribution, confirming that X-SAM successfully locates flatter minima by suppressing the dominant eigenvalue.

C.1. Evaluation on Test Loss and Accuracy

We evaluated the test accuracy and test loss of various methods using the ResNet-50, WideResNet, and AlexNet architectures. The experimental results are presented in Figure 7. As indicated by the accuracy curves, X-SAM consistently outperforms SAM, GSAM, and other baseline methods across all tested architectures and datasets, demonstrating superior generalization capabilities. Notably, we observed an intriguing phenomenon during our experiments: although X-SAM achieves the highest test accuracy, its corresponding test loss is not always the lowest; in some instances, it is marginally higher than that of standard SAM or other variants.

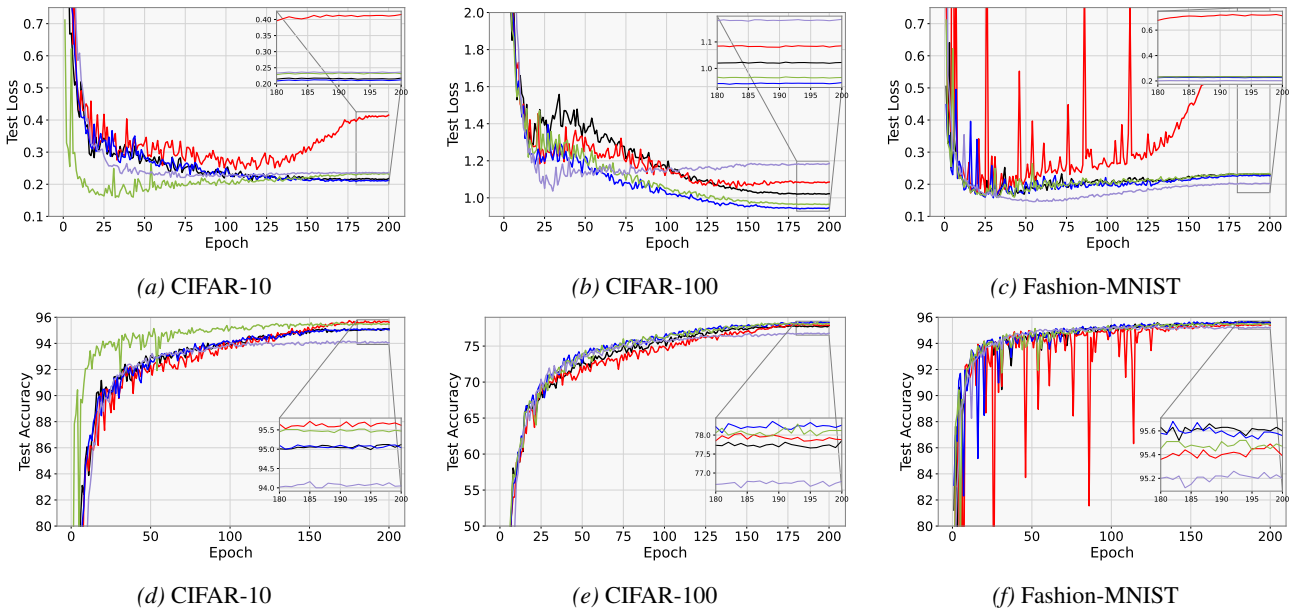
This phenomenon of "high accuracy accompanied by relatively higher loss" suggests that X-SAM does not merely pursue the minimization of cross-entropy loss (i.e., reducing the probability confidence error of predictions). Instead, it prioritizes the optimization of decision boundary robustness. A lower test loss often implies that the model is extremely "confident" (potentially over-confident) in its predictions, which can be an indicator of overfitting. Conversely, by explicitly penalizing high-curvature regions, X-SAM guides the optimization process into flatter minima. In such regions, the model exhibits more cautious predictions on misclassified samples, thereby avoiding extreme probability distributions on outliers. Consequently, the slightly elevated test loss actually reflects X-SAM's enhanced generalization ability and noise robustness, achieved while maintaining correct classification.



(a) CIFAR-10 *(b)* CIFAR-100 *(c)* Fashion-MNIST

(d) CIFAR-10 *(e)* CIFAR-100 *(f)* Fashion-MNIST

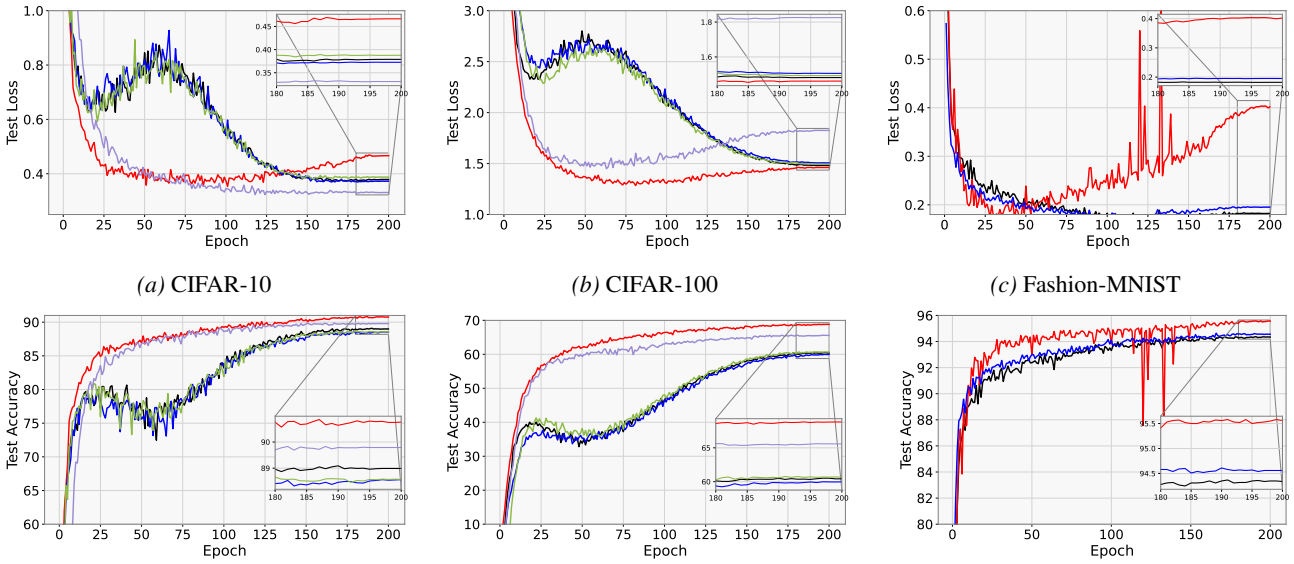
Figure 5. Learning curves for training ResNet-50 on CIFAR-10, CIFAR-100, and Fashion-MNIST. Figs 5a, 5b and 5c: Test loss vs. Epochs; Figs 5d, 5e and 5f: Test accuracy vs. Epochs.



(a) CIFAR-10 *(b)* CIFAR-100 *(c)* Fashion-MNIST

(d) CIFAR-10 *(e)* CIFAR-100 *(f)* Fashion-MNIST

Figure 6. Learning curves for training WideResNet on CIFAR-10, CIFAR-100, and Fashion-MNIST. Figs 6a, 6b and 6c: Test loss vs. Epochs; Figs 6d, 6e and 6f: Test accuracy vs. Epochs.



(a) CIFAR-10 *(b)* CIFAR-100 *(c)* Fashion-MNIST

(d) CIFAR-10 *(e)* CIFAR-100 *(f)* Fashion-MNIST

Figure 7. Learning curves for training AlexNet on CIFAR-10, CIFAR-100, and Fashion-MNIST. Figs 7a, 7b and 7c: Test loss vs. Epochs; Figs 7d, 7e and 7f: Test accuracy vs. Epochs.

C.2. Robustness to Perturbation Radius

To investigate the effect of X-SAM on SAM's sensitivity to the perturbation radius ρ, we conducted experiments on the CIFAR-10 dataset using the ResNet-18 architecture. Specifically, we compared the test performance of SAM and X-SAM under different perturbation radii $\rho \in \{0.05, 0.1, 0.15, 0.2\}$, repeating each setting three times to report the averaged test accuracy. The figure 8 shows that as the perturbation radius increases, SAM's test accuracy initially decreases and then slightly recovers, whereas X-SAM maintains a generally stable performance. Even at $\rho = 0.1$, X-SAM preserves consistent generalization. These findings indicate that X-SAM alleviates SAM's sensitivity to the perturbation radius, enabling more stable training dynamics and consistent generalization across a wide range of ρ values.

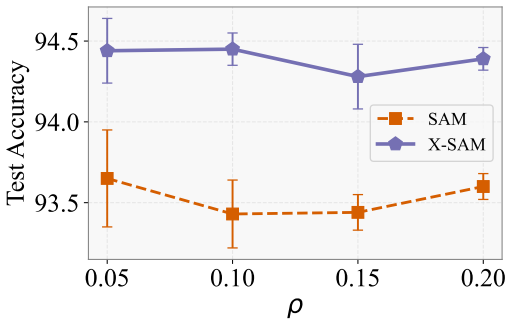


Figure 8. Comparison of test accuracy under different values of ρ on CIFAR-10 with the ResNet-18 model

C.3. Hyperparameter Sensitivity Investigation of alpha

We explore how the hyperparameter α influences the performance of X-SAM by running experiments on the CIFAR-100 dataset with the ResNet-18 architecture. Different values of α are evaluated, as illustrated in Figure 9. The results indicate that when $\alpha = 0.2$, the model achieves the highest test accuracy among all tested settings, showing a 1.07% improvement over standard SAM. This suggests that selecting an appropriate α can significantly enhance performance. Moreover, the test accuracy remains relatively stable when α varies within the range $[0.1, 0.4]$, indicating that α is a fairly robust hyperparameter.

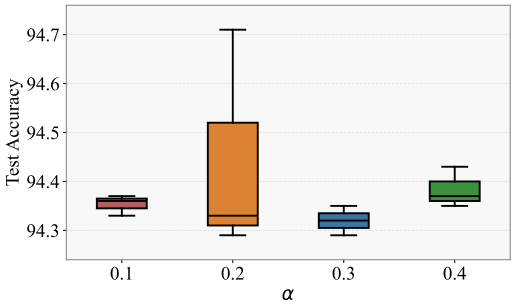


Figure 9. Test accuracy of ResNet-18 when training on CIFAR-10 under different values of α for X-SAM.

C.4. Hessian Spectrum Investigation

We analyze the impact of training with SAM and X-SAM on the Hessian spectrum of the model. The experiment is conducted on a ResNet-18 model trained on the CIFAR-100 dataset, where the top k eigenvalues of the Hessian matrix $\nabla^2 f(w)$ are computed at the second epoch and visualized. The results, as shown in Figure 10, show that compared with SAM, X-SAM consistently produces a smaller largest eigenvalue and a smaller trace, with a higher concentration of eigenvalues near zero. This indicates that X-SAM guides the optimization towards flatter regions of the loss surface, confirming the effectiveness of the proposed gradient correction mechanism and explaining why Eigen-SAM achieves better generalization performance than SAM.

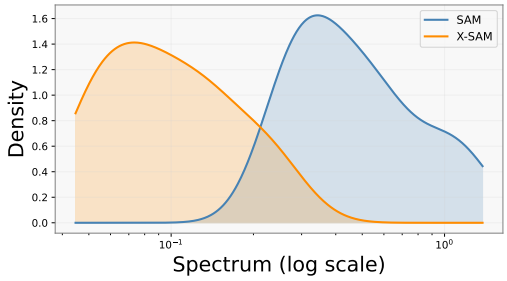


Figure 10. Hessian spectrum at the end of training ResNet-18 on CIFAR-100.

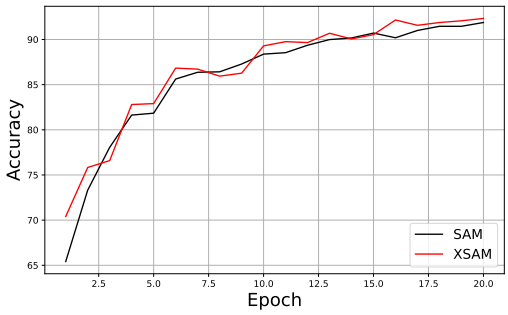


Figure 11. Test accuracy of the ViT-Tiny model fine-tuned on the CIFAR-10 dataset.

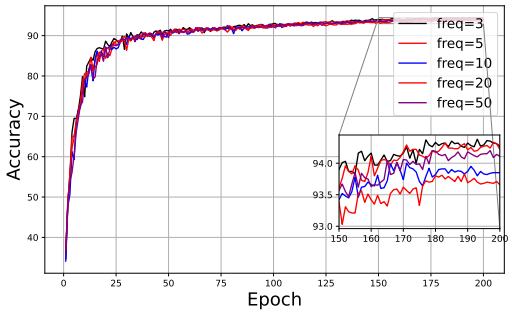


Figure 12. Test accuracy of ResNet-18 on CIFAR-10 with different eigenvector update intervals.

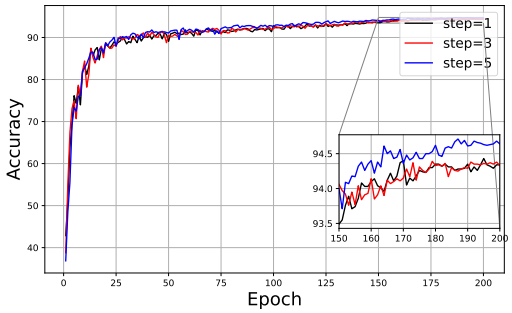


Figure 13. Test accuracy of ResNet-18 on CIFAR-10 with different numbers of power iterations.

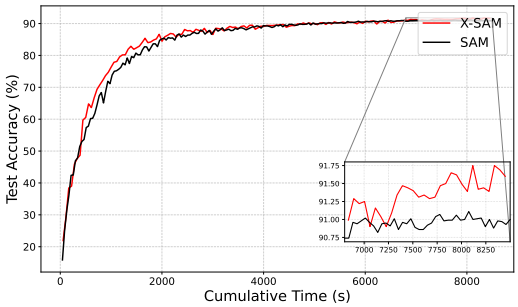


Figure 14. Test accuracy plotted against cumulative training time for ResNet-50 on the CIFAR-10 dataset.

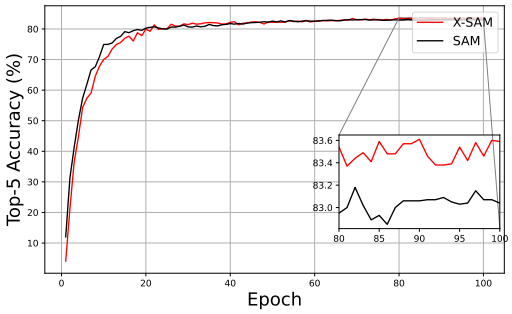


Figure 15. Top-5 Accuracy of ResNet-34 on the Tiny ImageNet dataset.

Datasets	Methods	Hyperparameters				
		LR	Weight Decay	ρ	α	γ
Model: ResNet-18						
CIFAR-10	WSAM	0.05	1e-3	0.20	–	0.88
	SAM	0.10	5e-5	0.05	–	–
	GSAM	0.10	5e-5	0.05	0.02	–
	Eigen-SAM	0.10	5e-5	0.05	0.20	–
	X-SAM	0.10	5e-5	0.05	0.20	–
CIFAR-100	WSAM	0.05	1e-3	0.20	–	0.82
	SAM	0.10	5e-5	0.10	–	–
	GSAM	0.10	5e-5	0.10	0.03	–
	Eigen-SAM	0.10	5e-5	0.10	0.20	–
	X-SAM	0.10	5e-5	0.10	0.20	–
Fashion-MNIST	WSAM	0.05	1e-3	0.20	–	0.88
	SAM	0.10	5e-5	0.05	–	–
	GSAM	0.10	5e-5	0.05	0.02	–
	Eigen-SAM	0.10	5e-5	0.05	0.20	–
	X-SAM	0.10	5e-5	0.05	0.20	–
Model: ResNet-50						
CIFAR-10	WSAM	0.10	1e-3	0.20	–	0.88
	SAM	0.10	5e-5	0.05	–	–
	GSAM	0.10	5e-5	0.05	0.01	–
	Eigen-SAM	0.10	5e-5	0.05	0.20	–
	X-SAM	0.10	5e-5	0.05	0.20	–
CIFAR-100	WSAM	0.10	1e-3	0.20	–	0.94
	SAM	0.10	5e-5	0.10	–	–
	GSAM	0.10	5e-5	0.10	0.20	–
	Eigen-SAM	0.10	5e-5	0.10	0.20	–
	X-SAM	0.10	5e-5	0.10	0.20	–
Fashion-MNIST	WSAM	0.10	1e-3	0.20	–	0.88
	SAM	0.10	5e-5	0.05	–	–
	GSAM	0.10	5e-5	0.05	0.01	–
	Eigen-SAM	0.10	5e-5	0.05	0.20	–
	X-SAM	0.10	5e-5	0.05	0.20	–

Table 2. Hyperparameters to reproduce experimental results on WideResNet and AlexNet.

Datasets	**Methods**	**Hyperparameters**				
		LR	Weight Decay	ρ	α	γ
Model: WideResNet						
CIFAR-10	WSAM	0.10	1e-3	0.20	–	0.88
	SAM	0.10	5e-5	0.05	–	–
	GSAM	0.10	5e-5	0.05	0.01	–
	Eigen-SAM	0.10	5e-5	0.05	0.20	–
	X-SAM	0.10	5e-5	0.05	0.20	–
CIFAR-100	WSAM	0.05	1e-3	0.20	–	0.94
	SAM	0.10	5e-5	0.10	–	–
	GSAM	0.10	5e-5	0.10	0.20	–
	Eigen-SAM	0.10	5e-5	0.10	0.20	–
	X-SAM	0.10	5e-5	0.10	0.20	–
Fashion-MNIST	WSAM	0.10	1e-3	0.20	–	0.88
	SAM	0.10	5e-5	0.05	–	–
	GSAM	0.10	5e-5	0.05	0.02	–
	Eigen-SAM	0.10	5e-5	0.05	0.20	–
	X-SAM	0.10	5e-5	0.05	0.20	–
Model: AlexNet						
CIFAR-10	WSAM	0.05	1e-3	0.20	–	0.88
	SAM	0.10	5e-5	0.05	–	–
	GSAM	0.10	5e-5	0.05	0.02	–
	Eigen-SAM	0.10	5e-5	0.05	0.20	–
	X-SAM	0.10	5e-5	0.05	0.20	–
CIFAR-100	WSAM	0.10	1e-3	0.20	–	0.82
	SAM	0.10	5e-5	0.10	–	–
	GSAM	0.10	5e-5	0.10	0.03	–
	Eigen-SAM	0.10	5e-5	0.10	0.20	–
	X-SAM	0.10	5e-5	0.10	0.20	–
Fashion-MNIST	WSAM	0.05	1e-3	0.20	–	0.88
	SAM	0.10	5e-5	0.05	–	–
	GSAM	0.10	5e-5	0.05	0.01	–
	Eigen-SAM	0.10	5e-5	0.05	0.20	–
	X-SAM	0.10	5e-5	0.05	0.20	–